

Block 7

Financial Stability

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UNIT 24 MACROECONOMICS, FINANCE AND BUSINESS CYCLES

Structure

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24.0 OBJECTIVES

After reading this Unit, you should be able to:

- identify macroeconomic issues in an economy;
- identify the examples of business cycles;
- explain the functioning of the financial system;
- identify asymmetric information and its effects;
- define financial crisis; and
- explain the linkages between the financial crisis and working of the economy.

24.1 INTRODUCTION

As discussed in previous blocks, the financial system plays an important role in the economy and has always been an interesting subject for economists and policymakers. In the present-day world, the financial sector of a country

is very important not only for advancement or growth but also for the stability of the economy. We have discussed in Unit 1 that the financial system includes three key elements: (i) financial institutions, (ii) financial markets, and (iii) financial instruments. The major function of the financial system in an economy is to work as an intermediary between lenders (or, savers) and borrowers (or, investors). Governments, corporations/business firms and individuals borrow from the financial markets to fulfil their objectives and invest their savings in financial instruments to grow their wealth. These decisions, in turn, affect macroeconomic variables like aggregate demand, output, employment, and prices. Macroeconomic variables affect financial conditions in a country and the developments in the financial sector are important for capital formation which is a prerequisite for the growth and development of the economy. However, we have also seen that at times, the developments in the financial sector leads to macroeconomic instability. The memory of the financial crisis of 2007-08 in the USA which spread across the globe within a short span of time, which is also known as the Great Financial Crisis or the Great Crash, and its after-effects on the world economy are very fresh. In this unit, we will extend our understanding of financial systems and examine the interconnection between Macroeconomics, finance and business cycles.

24.2 MACROECONOMICS AND BUSINESS CYCLES

In simple terms, macroeconomics is the study of the economy as a whole, i.e., how the economy works. It is focused on important issues like employment, inflation, output or GDP, growth rate, exchange rate, balance of payments, and business cycles. In short, we can say that it deals with the most important economic issues and problems of society. These important issues are dealt with mainly by two questions. One, how is the GDP of a country determined in the short run and the long run? Second, how is GDP (or the process of GDP determination) affected by macroeconomic variables such as consumption, investment, inflation and unemployment?

One of the major factors responsible for the growth of an economy is capital formation. Over the years, many financial innovations have taken place which have helped not only in building a capital base of the economy but also in the expansion of markets and wealth. But, the process or path of economic growth is not smooth or stable. Economies of different countries have experienced wide fluctuations in variables like employment, output, and prices over the past three centuries. Various kinds of explanations have been provided for economic fluctuations or business cycles ranging from the lack of aggregate demand to real business cycle models. Depending upon the kind of explanation, solutions for stabilising the economy have also been suggested. It has also been experienced that business cycles are often related to problems in the financial sector of the economy.

Business Cycles: It refers to ups and downs in economic activities namely, output, unemployment, and income among other things. The business cycle is an indication of how an economy is doing. A business cycle has four different stages predominantly. The first stage of a business cycle is an economic expansion. It is when the investment is increasing due to lower interest rates, output increases, and employment increases. The second stage is when the economic activities are at their peak and thus, the economy is doing extremely well. Often, it is considered that if there is a boom there is going to be a burst eventually. So, in short, in the first stage of a business cycle, the output is high, demand is high, income is high, unemployment is less, consumption is high, and investment is high. The second stage is when the economic activities are at their peak and thus, the economy is doing extremely well. In the third stage of a business cycle, aggregate demand goes down, the output starts to slow down, investment slows down, employment falls, income falls, and thus consumption also decreases. In the fourth stage of the business cycle, the condition is worse and a depression follows. Aggregate demand further decreases, investment decreases, production decreases again, and more layoffs happen.

The business cycle acts as a vicious cycle. Starting from a boom to a recession and then to a depression. One thing leads to another. For example, a lack of demand in one sector might lead to a reduction in investment and production which leads to loss of employment, a reduction in income and again a decrease in aggregate demand.

24.2.1 Case Study: The Great Depression (1929-1941)

In the 1920s the growth rate in the US was high (4.7%) and the unemployment rate was very low (3.2%). There was an exorbitant growth in wealth. With this high wealth, the investment by the public increased in the stock market. The regulations were weak in the stock market which had led to higher speculation in the market. Eventually, the stock market crashed in 1929. On October 24, 1929, the market opened down by 11% from the previous day. This is known as the 'Black Thursday'. In the following days, the market fell further. From 1929 to 1932, the stock market decreased by 85%. During the peak of the great depression, the unemployment rate was as high as 24%. A lot of blame was put on banks that had advanced too many loans by decreasing their reserves gradually. It was coupled with the expansionary monetary policy (increased money supply) which led to increased levels of borrowing. After the stock market crash instead of decreasing the rate of interest the Federal Reserve did the opposite and the rate of interest was increased. The approach was that the troubled banks would fail and a strong system would emerge. The increase in interest rate led to deflation (the opposite of inflation, prices decrease). At the same time higher tariff was imposed on agricultural imports to protect agriculture in the US which further contributed to the great depression. In 1933, President Roosevelt increased government spending but it was too late by then. Within the four years of the recession, the GDP of the US decreased by 29%.

Unemployment was at the highest (24%) level in 1933 when 1 out of every 4 people was unemployed. 30% of the banks failed. The standard of living decreased. Many factories had shut down. Rural poverty increased rapidly. The birth rate declined, and women's participation in the workforce increased. The great depression eventually ended in 1941.

24.3 FINANCE AND ECONOMY

We are generally faced with two kinds of situations. First, all of our income is not consumed and some of it is saved. For these savings, we look for options to earn some return, i.e., to invest our savings productively. Second, some people have business ideas or they have to do some transaction but do not have sufficient funds, i.e., they need funds. In macroeconomics, savings and investment are very important concepts and help in understanding macroeconomic problems like growth, inflation and unemployment and their solutions. Investment in physical and human capital is very important for a country's long-run growth. At the same time, it affects the macroeconomic stability, inflation and unemployment, in the economy. Savings are the most important component of funding investment, but every increase in savings may not be good for the health of the economy. It can lead to a shortage of aggregate demand in the economy. Moreover, it is also important how these savings are allocated to different investment options. Some options can be simply better than others in terms of productivity, outcome and impact on employment, inflation and other economic variables. The financial system of a country is generally helpful in providing solutions for the issues mentioned above. It provides productive investment options where savings can be invested and at the same time, the people in need get the opportunity to borrow the desired funds. It provides an efficient way to allocate the funds among different uses.

The financial sector is considered to be different from the real economy. The real economy produces goods and services with the help of the factors of production, land, labour, and capital. It provides us with consumption goods like food, furniture, clothes, and capital goods like plant, machinery, transportation, etc. What happens in the real economy determines an economy's capacity to produce and consume in the long run, and also affects several variables like inflation, unemployment, and distribution of income and wealth which in turn affect the economy itself. On the other hand, the financial sector provides the mechanisms and channels through which the lenders and borrowers coordinate in the market and facilitate transactions in the real economy. Efficient working of the financial system is very much desirable to increase the productivity in the economy, so that, the funds can be directed to those activities which are most productive. It is usually thought to be limited to the movement of funds in the economy to help the real economy. We can see that employment in this sector is increasing day by day. The financial system of a country not only channels the funds in the

right direction but in doing so generates well-paid jobs and builds wealth for many which in turn would have an impact on activities like consumption and production. It is often a matter of debate whether the financial sector helps in the development of the economy or if it is a result of economic development, both of these arguments are true.

24.3.1 Financial System

The financial sector, today, is a very important sector for almost all economies. The major function of this sector is to provide channels for the movement of funds between borrowers and lenders. Lenders and borrowers can be individuals, households, business firms, governments and foreigners. Lenders and borrowers can interact through organised or formal financial markets or outside these markets. The transactions done outside the organised financial markets are, generally, riskier and costlier as compared to the formal markets. The financial system of a country helps in reducing the transaction costs and risks involved in financial transactions. As a country develops, its financial market also develops along with it and the proportion of the transactions outside formal financial markets declines with economic development.

The financial system includes financial institutions like banks, insurance companies, mutual funds etc. and financial markets like stock markets, bond markets etc. In financial markets, lenders can directly provide funds directly to the borrowers. For example, suppose Reliance Industries wants to expand its business and for that purpose needs some funds. It can directly sell bonds to individuals in the bond market or it can bring an IPO (initial public offering) in the stock market which can be subscribed by individuals, banks, and companies. This is **direct finance**. In this case, the individuals or the companies who have savings are directly lending their money to the borrower or buying stocks of a company. This requires a lot of information and expertise to deal directly with the financial markets. On the other hand, instead of lending directly to the borrower, an individual can do it through some intermediary. For example, I have some savings and I want to invest these savings profitably. I can deposit my money in a bank, I can invest in a mutual fund, I can buy some insurance, or I can participate in some pension scheme. Financial institutions like banks, mutual funds, insurance funds or pension funds receive money from the lenders and lend it to the borrowers. This is **indirect finance**.

In financial markets, there are different kinds of risks present for individuals, firms, and financial intermediaries. There is a risk that the deposits or the credit given may not be paid back partially or fully, known as **credit or default risk**. Financial intermediaries including banks may find themselves unable to meet the liquidity demands due to sudden and unexpected increases in the demand for liability payments or withdrawal of deposits, known as **liquidity risk**. Volatility in the interest rates and asset prices can deteriorate the financial health or the balance sheets, known as **interest rate and price**

risk. Volatility in the exchange rate affects the value of assets and liabilities held in foreign currency and may worsen the balance sheets, known as the **exchange rate risk**. Due to one or more risks mentioned above, the value of assets of financial intermediaries, or firms and individuals, may fall below liabilities leading to **insolvency**. One of the important reasons for the risk is **asymmetric information**. One of the important aspects of the functions of financial intermediaries is to assess and manage these risks.

24.3.2 Asymmetric Information, Adverse Selection and Moral Hazard

When, in a transaction, one party does not have the same information as the other party it is known as **asymmetric information**. For example, suppose there is a firm which is planning to undertake a new project and for this project firm borrows from the market. The firm has different assessments for the risks involved in the project, and prospects of the future return. But, while approaching the lenders or investors this firm may not reveal all the information related to the project it has. In this case, the lenders/investors would have less information than the borrower.

It happens at times that we try to hide some information which may impede our chances to borrow from the market. Suppose, an individual or a firm is willing to undertake a very risky project and lenders or investors may not be willing to take that kind of risk. To get the necessary funding the borrower may hide facts or information about the risk and make best efforts to get the loan, and may even offer to pay a higher interest rate. If the lenders/investors, pursued by the borrower, invest in the project without knowing the risk it creates **an adverse selection problem**. Due to the risky nature of the project, the borrower may not be able to honour its liabilities, i.e., the risk of default. There is another problem that arises because of asymmetric information which is known as **moral hazard**. This may occur irrespective of whether there is an adverse selection or not; this takes place after the financial transaction has occurred. A borrower may divert the borrowed funds to some risky project or activity. For example, the firm may put its investment in risky bonds that might pay or not pay. If these risky bonds pay then the bank will get its money back and if not, then the firm will not be able to pay back and most likely will default. Another simple example of moral hazard is that suppose you have taken insurance for losses due to fire. After taking the insurance you may decide not to put in place an efficiently working fire safety system to cut down on costs.

24.3.3 Case Study: Satyam Computers Scandal

Satyam Computer Services Limited (popularly known as Satyam Computers), a company established in 1987, was the fourth biggest Indian company in the IT sector in the 2000s. It was one of the finest examples of India's growth story. From 2003 to 2008, the annual growth rate of the company was 35 per cent. In January 2009, the Chairman and the founder of the company confessed that he had been manipulating the accounts of the

company. He inflated the balance sheet and showed the profits and cash which never existed. This false growth also inflated the stock prices of the company. This was done to obtain cheaper loans which were diverted to the real estate. The company's auditors turned a blind eye to the wrongdoings. The company lost its customers, stock prices plumped, and investors lost billions. The company was acquired by Tech Mahindra in April 2009. In Indian corporate history, the Satyam scandal is a very fine example of corporate fraud. This highlights the problem of asymmetric information that led to adverse selection due to false information and moral hazard on the part of managers and auditors. This episode also highlights the importance of prudent regulation that would help in decreasing the information asymmetry and promote transparency.

Check Your Progress 1

- 1) Discuss different stages of business cycles.

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- 2) How important is the concept of asymmetric information for the financial sector?

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- 3) What are the different kinds of risk associated with financial transactions?

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24.4 FINANCIAL CRISIS

The financial crisis can be understood as an upset in the financial markets in the form of wide fluctuations in asset prices like stocks or real estate. It is associated with fluctuations in the flow of credit and disruptions in the working of financial intermediaries and markets. Financial stability helps in better assessment and management of financial risk and thus helps in the efficient allocation of resources which in turn, promotes economic growth. Although, with financial innovations, the understanding of risk and its

management has also improved it is not possible to avoid risk completely because of the uncertainty.

The financial system is like a complex web of interactions between different players. For example, person A has invested money in Mutual Fund X, this mutual fund has invested money in another investment fund which has further invested money in a bank that has given a loan to a corporate firm. You have a financial claim over a part of the mutual fund. The amount of money you can get by selling your holding of the mutual fund depends upon many things. First, the price or value of the fund may be affected by the conditions of the market. Second, the value of underlying assets may further be a function of different variables. And third, if the mutual fund in question has sufficient liquidity. In financial markets, whether the contract between two parties will be honoured or not depends upon other parties who are linked to this contract by a chain of financial transactions. A disturbance at any point in this chain of transactions can cause financial turbulence in the market and it can be a reason for a **financial crisis**. Speculative activities in the financial markets along with asymmetric information are often the sources of financial fluctuations. There may be several other factors that may start or may increase the intensity of a financial crisis.

It is not so that all the financial crises are the same in the nature and magnitude but they have some common elements. Usually, a financial crisis starts with an asset price boom. These elements are described in the following paragraphs:

Asset Price Booms and Busts: At times, the prices of financial assets like stocks or real estate properties increase by a large amount over their **fundamental values**. This is known as **the asset-price bubble**. Often, this is led by the optimism of the investors about the future value of the asset. This may also happen because of a significant increase in the availability of credit. With the easy availability of credit, investors may increase their investment in an asset they expect to give very good returns in the future. This kind of increase can be termed a **credit boom** or **credit bubble**. The optimism about the asset prices and credit boom can have a feedback effect on each other, which may further intensify the asset-price bubble. The asset price booms cannot last forever, even a small shock in the market can lead to the bust of the bubble. For example, if investors start to feel the future rate of return for the asset may be lower, they may start to withdraw their investment which will lead to a decline in the asset prices.

Insolvencies of the financial institutions: At times, the loans given by the banks are secured by the collateral or mortgage of the asset for which the loan has been provided. When the prices of such assets decline, it will create problems for the lending banks. In case of defaults by the borrowers, the banks will not be able to recover their loans. This problem is further aggravated by the **leveraging** by the lending institutions. In an optimistic environment, to increase the amount of loans/credit banks often borrow from other financial institutions. The borrowed funds increase banks' capacity to

lend. This process is known as leveraging. But in case of asset-price bust, the value of mortgage declines and banks are unable to recover their loans and thus, fall short of meeting their payment obligations. In this situation, the capital of the banks falls and they reduce their lending. Which may lead to a further decrease in asset prices. The value of the assets of the banks declines below the value of liabilities which ultimately leads to insolvency.

Banking Failure (or Bank Runs): As the problems of the banks increase and banks start to go bankrupt, the depositors who do not want to lose their money, line up to withdraw their money from the banks. Though, some amount in deposits is insured by the governments, generally, it is very small. Banks do not have sufficient balances and they start to sell their assets. Asset prices are already low and it is hard to find buyers in the market. The selling of assets by the banks may lead to a rapid decline in asset prices. This is also known as a **fire sale**. Uncertainty about the health of the banks in the economy increases and people lose confidence in the safety of their deposits. This may lead to a full-fledged banking failure in the economy, i.e., several banks failing in a short duration.

Credit Shortage and Debt Deflation: When some banks fail, capital decreases as it becomes harder to get new capital as a result of low confidence in the market. Banks cut down on lending, and it becomes harder for the firms, especially the smaller and medium-sized firms, to get loans from the banks. Consumer credit also goes down as scrutiny by the banks increases. With the increase in the difficulties of the financial sector, the flow of credit declines in the economy. Decreases in the quantum of credit not only affect the new investment opportunities but also the operations of the firm which rely on debt for running their operation. A decrease in consumer credit leads to a decline in consumption demand. This leads to a decline in the economic activities in the country, a **recession**, and prices fall which increases the problems for the business firms. With the decrease in the prices, in real terms, the debt of the firms increases, known as **debt deflation**. Prices and incomes fall while the debt burden increases which leads to insolvency of the firms. This generates a kind of vicious circle. Deterioration in the health of the financial sector creates a shortage of credit that leads to an economic downturn which in turn decreases the profitability and stock prices of the firms. This may lead to insolvencies in the economy further deteriorating the health of the financial sector. In a financial crisis, several financial and non-financial firms collapse.

24.4.1 Case Study: The Great Recession (2007-2009)

From the great depression to the great recession, there were many recessionary periods in between. Out of these 2007 was the worst. Before this recession, there was the dot.com crisis in the US which was a crisis in the IT industry. To give a boost to this sector the US Federal Reserve decreased the rate of interest. Instead, this investment went to the real estate sector in the US. Soft loans were given for real estate properties. Multiple loans were advanced on the same mortgage. People who had poor credit history were

also given loans. Banks who had these mortgages advanced financial instruments (shares, bonds, futures etc.) could not recover their dues. These financial instruments were even purchased by people sitting in other countries. There was a bubble in the real estate which was likely to burst. Ultimately, the prices in real estate started to fall. People defaulted on loans. Looking at the panic, people who had bought securities based on these mortgages started to exit. So, on one hand, people did not pay loans to banks and on the other hand the prices of their mortgage-based securities decreased. It was a double-edged sword for the banking system. The NPAs increased drastically. Major financial institutions collapsed in 2008. Lehman Brothers which was the 4th largest bank at that time, collapsed in September 2008. Ultimately the recession had spread to another sector too. Initially, India did not see any significant impact but eventually, the impact could be seen through the financial sector and decreased demand for Indian exports by the US and Europe. As a result of this recession, output decreased by 4.3%. Banks failed; housing prices declined by 31.8 per cent; unemployment peaked at 10% in October 2009 while it was 5% in December 2007; public debt increased, and protectionism increased.

24.5 FINANCIAL CRISES AND ECONOMIC CRISES

The financial system is an integral part of the economy and, both in good times and bad times, it amplifies economic activities. As financial development helps in economic growth a financial crisis worsens the economic problems. It is not necessary that a financial crisis would lead to an economic crisis; an economic crisis may also be a cause for the financial crisis. Finance is a key factor in capital formation and business activities.

When finance deteriorates it has a severe impact on economic activities. As we have discussed, during a financial crisis asset prices and price levels and at the same time, an increase in interest rates cause a decrease in the profitability of the firms. Credit crunch and a decline in profitability lead to a decrease in overall economic activities. The defaults and insolvency in non-financial firms would increase.

As a financial crisis looms a country's capital may fly to other countries in search of safer investment opportunities. It would have a twofold impact on the country. One, it would further deteriorate the credit crunch in the economy. Second, with a flight of capital exchange rate will be affected adversely which will impact the trade. Trade disturbance would aggravate the economic turbulences.

An economic crisis can also create financial crisis. An economic downturn would decrease the sales and profitability of the firms which would lead to an increase in defaults in the repayments of loans they have borrowed from financial institutions. This would hurt the balance sheets of the banks and other financial institutions may further lead to a financial crisis.

Generally, a recession that is associated with a financial crisis lasts longer. It leads to a greater decrease in output and employment; one, directly through the collapse of the financial sector and second, through the consequential economic downturn. Financial crises also have a greater implication for fiscal outlays as governments have to focus on saving the financial institutions along with improving the aggregate demand in the economy.

24.6 POLICY RESPONSES TO A CRISIS

Now, that we know the adverse impacts of a financial crisis or an economic crisis the question arises what can be done? In this section, we will discuss in detail the macroeconomic policies to tackle a crisis. A single policy intervention, like monetary or fiscal, or both, is generally not sufficient for stability in the economy. A combination of different kinds of policy measures is required. The following policy measures can be used to fight a crisis.

Fiscal Policy: We know that in a crisis, financial or economic, unemployment increases, and output and aggregate demand decrease. Policymakers can use fiscal expansion, in the form of a tax cut or an increase in government spending, to increase aggregate demand in the economy. A tax cut would lead to an increase in disposable income in the economy, while, an increase in government spending can come in the form of expenditure on infrastructure, stimulus to industry, social infrastructure (health and education), and social welfare schemes. The magnitude of fiscal intervention that is required to counter the cyclical downturn depends upon the seriousness of the crisis. The bigger the crisis, the larger should be the fiscal expansion.

Monetary Policy: Like fiscal expansion, monetary policy can also be used to increase aggregate demand in the economy which would help in improving employment and output. A monetary expansion generally comes in the form of a reduction in interest rates and an increase in the liquidity or money supply in the economy. A decrease in interest rates would lead to a reduction in the cost of borrowing, and an increase in liquidity would help in easing the credit crunch in the economy.

Both monetary and fiscal policies have their limitations. A deep crisis may require a drastic decrease in the rate of interest but it cannot go below zero. Moreover, monetary policy is not useful in a condition of a liquidity trap. Fiscal expansion may hurt the interest rates and prices in the economy. In such situations, monetary and fiscal policy should be supplemented by different regulations directly addressing the issues related to asymmetric information, adverse selection, and moral hazard.

Strengthening of Regulatory Framework: Financial crisis is generally a result of asset-price bubbles. Policymakers should strengthen the regulatory and monitoring process of the financial sector. This requires continuous

supervision and evaluation of financial institutions and financial instruments. Financial innovations take place continuously in the market, in the form of new products and expansion of the market, which creates new challenges not only for the financial institutions and the markets but also the policymakers and regulators. This creates a need for continuous improvement in the regulatory framework.

Check Your Progress 2

- 1) Explain asset-price boom and bust with a suitable example.

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- 2) What do you understand by fire sale?

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- 3) How do financial and economic factors interact during a crisis?

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- 4) What policy options can be used to fight a recession?

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24.7 LET US SUM UP

The financial sector is very important for the overall growth of the economy. The major function of this sector is to provide channels for the movement of funds between borrowers and lenders. Speculative activities in the financial sector have major implications for the business cycles. The recession of 2007-08 better known as the subprime mortgage crisis, provides a very good example of this. Over the years many financial innovations have taken place which has helped not only in building capital base but also in expansion of

markets and wealth across the countries. However, the process or path of economic growth is not smooth or stable. Economies of different countries have experienced wide fluctuations in variables like employment, output, and prices over the past three centuries. The presence of asymmetric information, adverse selection better known as the subprime mortgage crisis, and moral hazard can also contribute to the crises. As financial development helps in economic growth a financial crisis worsens the economic problems. Booms in asset prices, fuelled by credit growth and positive expectations, followed by busts are generally the cause of the financial crisis. It is not necessary that a financial crisis would lead to an economic crisis; an economic crisis may also be a cause for the financial crisis. Finance is a key factor in capital formation and business activities. The policy response can be in terms of fiscal policy, monetary policy and regulatory reforms.

24.8 KEYWORDS

Business Cycle	: Refers to ups and downs in economic activities namely, output, unemployment, and income among other things.
Real Economy	: The real economy produces goods and services with the help of the factors of production, land, labour, and capital. It provides us with consumption goods like food, furniture, clothes, and capital goods like plant, machinery, transportation etc.
Asymmetric Information:	When in a transaction, one party does not have the same information as the other party it is known as asymmetric information.
Adverse Selection	: When the lenders/investors, pursued by the borrower, invest in the project without knowing the risk.
Moral Hazard	: This takes place after the financial transaction has occurred. A borrower may divert the borrowed funds to some risky project or activity.
Credit Boom	: With the easy availability of credit, investors may increase their investment in an asset they expect to give very good returns in the future.
Monetary Policy	: Increase or decrease in money supply by the central bank to target macroeconomic variables like unemployment, output, inflation etc.
Fiscal Policy	: Used for increase in output through Government purchases and taxes.

24.9 SOME USEFUL REFERENCES

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24.10 ANSWERS / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Economic expansion, economic activities are at their peak, and output starts to slow down and then downfall.
- 2) Refer to Sub-section 24.3.2
- 3) credit or default risk, liquidity risk, interest rate and price risk, and exchange rate risk.

Check Your Progress 2

- 1) At times, the prices of financial assets like stocks or real estate properties increase by a large amount over their **fundamental values**. This is known as **the asset-price bubble**.
- 2) The selling of assets by the banks may lead to a rapid decline in asset prices. This is also known as **a fire sale**.
- 3) Refer to Section 24.5
- 4) Fiscal Policy, Monetary Policy, Strengthening of Regulatory Framework

UNIT 25 EFFICIENT MARKET HYPOTHESIS

Structure

- 25.0 Objectives
- 25.1 Introduction
- 25.2 History of Efficient Market Hypothesis (EMH)
- 25.3 Efficient Market Hypothesis
- 25.4 Assumptions of EMH
- 25.5 EMH and Capital Asset Pricing Model (CAPM)
- 25.6 Assessment of Efficient Markets Hypothesis
- 25.7 Applications of the EMH
- 25.8 Applicability of the EMH in India
- 25.9 Let Us Sum Up
- 25.10 Key Words
- 25.11 Some Useful Books
- 25.12 Answers/ Hints to Check Your Progress Exercises

25.0 OBJECTIVES

After going through this Unit, you will be able to:

- explain the meaning of the Efficient Markets Hypothesis;
- describe the relationship of the Efficient Markets Hypothesis and Capital Asset Pricing Model (CAPM);
- critically examine the Efficient Markets Hypothesis;
- appreciate empirical techniques to test the Efficient Markets Hypothesis;
- apply the Efficient Markets Hypothesis to real-life data; and
- apply the Efficient Markets Hypothesis to data on Indian financial markets.

25.1 INTRODUCTION

A key idea in contemporary finance, the Efficient Markets Hypothesis (EMH) has revolutionised the way we think about the behaviour of financial markets. According to the hypothesis, financial markets are incredibly effective, and the prices of securities exchanged there always represent all

available information. To put it another way, market prices react quickly and precisely to any new information that enters the market, making it nearly hard for investors to continuously generate above-average profits by trading on public information.

The concept of EMH has been around for more than 50 years now. The hypothesis has been widely debated by scholars and practitioners due to its assumptions. Some scholars argue that EMH is flawed, while others maintain that it provides a realistic and accurate representation of the functioning of financial markets. Nevertheless, EMH has played a crucial role in shaping modern financial theory, and it remains a key subject of study for students and professionals in the field of finance.

The present unit aims to provide a comprehensive introduction to the Efficient Markets Hypothesis and its applications in the field of finance. We explore the origins and evolution of the theory, examine its key assumptions and implications, and review the empirical evidence supporting and challenging its validity. We also discuss various forms of market efficiency, such as weak, semi-strong, and strong efficiency, and how they relate to different types of information available to investors. Finally, we discuss the practical implications of EMH for investment decision-making, asset pricing, and portfolio management.

Whether you are a finance professional, a student, or simply interested in understanding how financial markets work, this unit will provide you with a solid foundation in the Efficient Markets Hypothesis and its importance in the world of finance.

25.2 HISTORY OF EFFICIENT MARKETS HYPOTHESIS

The Efficient Markets Hypothesis (EMH) is a theory in finance that states that financial markets are “informationally efficient,” meaning that asset prices reflect all available information at any given time. The basic idea behind the EMH is that in an efficient market, investors cannot consistently earn abnormal returns (i.e., returns that are higher than what would be expected given the level of risk they are taking on) by using any available information, because that information is already reflected in the market price.

The history of the efficient markets hypothesis dates back to the early 20th century when economists such as John Bates Clark and Alfred Marshall developed theories about the relationship between prices and information. However, the modern version of the EMH began to take shape in the 1950s and 1960s, when economists such as Eugene Fama, Paul Samuelson, and Burton Malkiel began to develop and refine the theory.

Eugene Fama is often credited with creating the modern version of the EMH in his 1965 paper “Random Walks in Stock Market Prices,” which argued

that stock prices follow a random walk and that it is impossible to consistently predict future stock prices based on past prices or any other available information. This paper, along with others by Fama and other economists, helped establish the EMH as a cornerstone of modern finance theory.

Since its inception, the EMH has been the subject of much debate and controversy. Some economists and investors argue that the EMH is an accurate representation of how financial markets work, while others believe that markets are not completely efficient and that it is possible to consistently earn abnormal returns through various investment strategies. Nonetheless, the EMH remains an important concept in finance and continues to influence the way that investors think about the relationship between information and asset prices.

25.3 EFFICIENT MARKETS HYPOTHESIS (EMH)

The Efficient Markets Hypothesis (EMH) is a theory in finance that suggests that financial markets are “efficient,” which means that asset prices fully reflect all available information at any given time. The hypothesis states that it is impossible to consistently achieve returns over average market returns because any new information that enters the market is immediately and fully reflected in the asset prices. In other words, the EMH suggests that it is impossible to consistently beat the market by using information that is already publicly available since the market has already used that information to arrive at the prevailing price. The theory argues that market players cannot earn higher returns than the market as a whole, except by taking on more risk. Eugene Fama (1970) has given three forms of the EMH which are as follows:

- 1) **Weak Form:** The weak form of EMH asserts that all historical price and volume data is already incorporated into current stock prices, making it impossible to profit from analyzing past trends or patterns. This means that technical analysis, which involves using charts and past market data to predict future prices, cannot generate consistent profits.
- 2) **Semi-Strong Form:** The semi-strong form of EMH asserts that all publicly available information is already reflected in stock prices, making it impossible to achieve abnormal returns through fundamental analysis. This means that researching and analyzing company financial statements, news articles, and economic data cannot lead to consistent profits.
- 3) **Strong Form:** The strong form of EMH asserts that all information, both public and private, is already reflected in stock prices, making it impossible to achieve abnormal returns through any form of analysis or insider information.

The following diagram summarises the Efficient Markets Hypothesis.

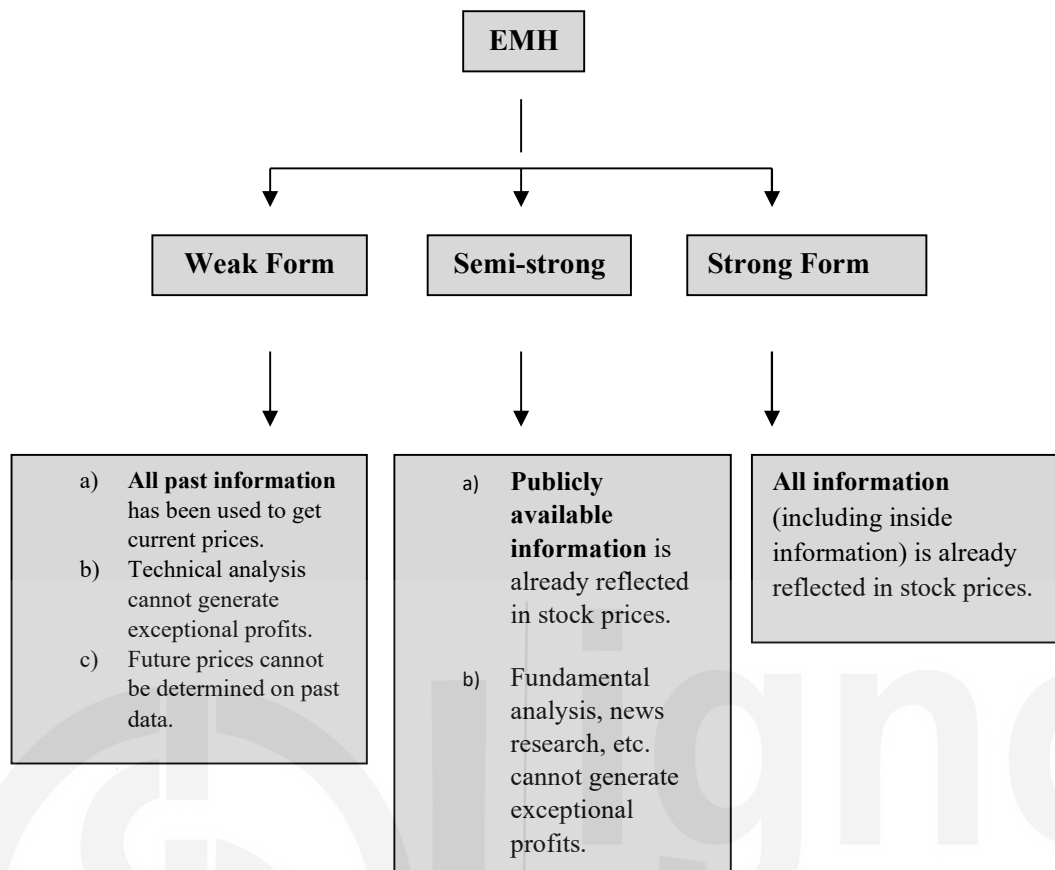


Fig. 25.1: Different Forms of Efficient Markets Hypothesis

While the efficient markets hypothesis is widely debated, it has influenced much of modern financial theory and has important implications for investors, financial analysts, and policymakers.

25.4 ASSUMPTIONS OF EMH

The Efficient Markets Hypothesis is based on the following assumptions:

- 1) **Rationality:** Investors are rational in the sense that they maximise their interests. They do their best.
- 2) **Information is freely available:** Every investor has easy accessibility to all market-related information without any restrictions. In other words, the information is available at zero cost.
- 3) **No asymmetry of information:** It means every investor has the same amount of information i.e., whatever information investor A has, then the same information is available to B also.
- 4) **Prices adjust quickly to new information:** The response time of the market to new information is much less. In other words, as new information is made available, then its effect can be seen in stock prices and indices very quickly.

- 5) **No individual investor can influence prices:** No investor, or group of investors, has the force to control prices and steer the market in a specific direction.
- 6) **No transaction costs:** Investors are not prevented from acting on their information by transaction costs or other trading obstacles.

Check Your Progress 1

- 1) What is the basic idea underlying the Efficient Markets Hypothesis?

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- 2) What are the various forms of the EMH? What do they imply?

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- 3) State the assumptions of the Efficient Markets Hypothesis.

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25.5 EMH AND CAPITAL ASSET PRICING MODEL (CAPM)

The Efficient Markets Hypothesis (EMH) and the Capital Asset Pricing Model (CAPM) are two important concepts in finance that are closely related to each other.

The Efficient Markets Hypothesis states that (i) financial markets are efficient, and (ii) all available information is reflected in asset prices. According to the EMH, it is impossible to consistently beat the market because all available information is already reflected in the asset prices. In other words, the market is already in equilibrium and all the stocks are valued at their intrinsic values. Therefore, a stock cannot perform better than the market.

The Capital Asset Pricing Model (CAPM) as we have already discussed in Unit 20 is an asset pricing model which was suggested by economists William F. Sharpe (1964) and Linter (1965). The CAPM states that the expected return of an asset (or a security portfolio or stock) is linearly and positively related to its systematic risk (the part of total risk which is beyond the control of the firm or the investor; it is determined by various economic, social and political factors). The CAPM predicts that the systematic risk of an

asset has the power to explain the expected return of an asset in a linear fashion. Sharpe and Linter established the following equation of the CAPM:

$$E(R_i) = R_f + [E(R_m) - R_f]\beta_i \quad \dots (25.1)$$

where $E(R_i)$ is the expected return of an asset i , R_f represents the rate of return of a risk-free asset (such as T-bills or Government securities), $E(R_m)$ is the expected return of a market portfolio (typically proxied by an index of a stock exchange such as SENSEX or NIFTY), and β_i is a measure of the systematic risk. Apart, the expression $E(R_m) - R_f$ is called market risk-premium.

The CAPM is based on the following main assumptions:

- a) investors are rational,
- b) they diversify their portfolios among various financial assets or instruments, and
- c) the market is efficient.

The point to be noted here is the assumption that “the market is efficient”. Thus, the CAPM states that the EMH holds good.

Specifically, the CAPM assumes that markets are efficient, which means that all relevant information is already incorporated into asset prices. In other words, the CAPM assumes that investors cannot earn higher returns by using publicly available information, which is a core principle of the EMH.

The CAPM is also used to test the efficiency of markets. If the CAPM holds, it would suggest that markets are efficient, because it implies that investors cannot earn higher returns by using publicly available information. However, if the CAPM is not able to explain returns on assets, it may suggest that markets are not efficient.

An Econometric Technique to Test the EMH Using the CAPM

Black, Jensen, and Scholes (1972) suggested a test statistic to test the CAPM. They used the following equation for each security i :

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + u_{it} \quad \dots (25.2)$$

The left-hand side of equation (25.2), that is, $(R_{it} - R_{ft})$ is called ‘the excess return of security i over the risk-free return’. The expression $(R_{mt} - R_{ft})$ on the right-hand side is called ‘market risk premium’ or the excess return of the market portfolio over the risk-free return. Thus, the above equation can be re-written as follows:

$$ER_{it} = \alpha_i + \beta_i ER_{mt} + u_{it} \quad \dots (25.3)$$

In equation (25.3) the term ER_{it} indicates excess returns of the i^{th} security and the term ER_{mt} indicates the average expected return in the market. The coefficient β indicates the systematic risk.

Black, Jensen, and Scholes (1972) suggest that if the CAPM is working, then α_i should not be statistically different from zero for all n securities. Thus, their null hypothesis is $\alpha_i = 0$. If the null hypothesis is rejected, that is, if $\alpha_i \neq 0$, then the market is not efficient in the light of the efficient markets hypothesis. If we plot equation (25.3) on graph paper, we obtain the characteristic line. If the EMH holds good, then the characteristic line should pass through the origin as shown in Fig. 25.2.

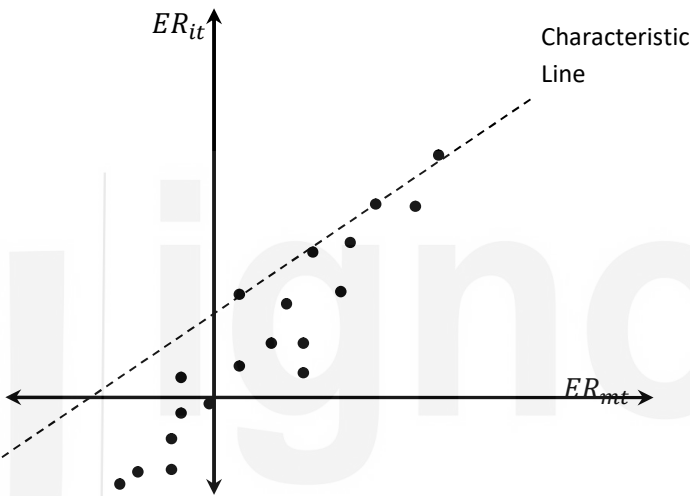


Fig. 25.2: Capital Asset Pricing Model

The overall conclusion that we draw is that the CAPM and the EMH are two closely related concepts in finance. The CAPM is built on the assumptions of the EMH, and the CAPM can be used to test the efficiency of markets.

Check Your Progress 2

- 1) What are the major inferences of the Capital Asset Pricing Model? How is it related to the Efficient Markets Hypothesis?
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- 2) Explain how the EMH can be tested by using the CAPM.
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25.6 ASSESSMENT OF EFFICIENT MARKETS HYPOTHESIS

The Efficient Markets Hypothesis (EMH) is a theory that suggests that financial markets are informationally efficient, meaning that asset prices reflect all available information at any given time. While this hypothesis has been widely accepted in the academic world, it has also received significant criticism from various scholars and practitioners in finance. Some of the main criticisms of the EMH are:

- a) **Behavioural finance:** We know that the EMH is based upon an assumption of the rationality of investors, but the studies conducted under behavioural finance argue that investors are not always rational, and their decisions are derived from their emotions, biases, or incomplete information, causing inefficiencies in the market. This means that market prices may not always reflect all available information, as suggested by the EMH.
- b) **Historical contradictions:** Some historical anomalies or contradictions cannot be explained by the EMH. For example, the momentum effect, where stocks that have performed well in the past tend to continue to perform well in the future, and this tendency contradicts the EMH which states that future prices cannot be predicted based on past prices.
- c) **Insider trading:** The existence of insider trading suggests that some market participants have access to information that is not available to the general public, indicating that the market may not be as efficient as the EMH suggests.
- d) **Market bubbles:** The EMH contends that the price of a security tends to its intrinsic value i.e., it is neither overvalued nor undervalued. But, the cases of market bubbles such as the dot-com bubble of the late 1990s and the housing bubble of the mid-2000s suggest otherwise. In both cases, asset prices soared far beyond their true intrinsic value, leading to a crash when the bubble eventually burst.
- e) **Information asymmetry:** The EMH assumes that all market players have equal information i.e., nobody has more information than the others. Do you think so? In reality, some participants may have more or better information than others. This creates an information asymmetry, which can lead to market inefficiencies.
- f) **Limited arbitrage:** The EMH argues that the arbitrage process in the market corrects these inefficiencies caused by over and under-valuation

of the stocks. However, in reality, arbitrage opportunities are often limited by factors such as transaction costs, liquidity constraints, and regulatory restrictions, which may prevent market prices from reflecting all available information.

So, the overall conclusion is that the above criticisms suggest that the EMH may not accurately describe the behaviour of financial markets and that market inefficiencies may be more common than the EMH suggests.

25.7 APPLICATIONS OF THE EMH

Here are some main applications of the EMH:

- a) **Index Funds:** The EMH suggests that it is impossible to consistently outperform the market by picking individual stocks. In other words, the market prices of the stock are already at their best values and one stock cannot perform better than the market return. Therefore, investors can choose to invest in index funds that track the performance of the overall market.
- b) **Passive Investing:** Passive investing means an investing strategy which aims to track the performance of a market index (such as Sensex or Nifty) instead of thinking that some stock will perform better than the market. The EMH supports the idea of passive investing, which involves buying and holding a diversified portfolio of stocks for the long term, rather than trying to time the market or pick individual stocks.
- c) **Rational Expectations Theory:** The EMH is closely related to the Rational Expectations Theory, which suggests that individuals make decisions based on all available information and that markets adjust rapidly to new information.
- d) **Academic Research:** The EMH has been the subject of a great deal of academic research, particularly in the fields of finance and economics. Researchers have used EMH as a framework to test various hypotheses about market behaviour.
- e) **Regulatory Policy:** The role of regulatory authorities like the Securities Exchange Board of India (SEBI) may get diluted by implication of the EMH. The EMH has implications for regulatory policy, as it suggests that attempts to regulate markets or prevent insider trading may be futile since all available information is already reflected in stock prices.
- f) **Investment Strategies:** The EMH suggests that traditional investment strategies, such as technical analysis and fundamental analysis, may be ineffective. Instead, investors may wish to focus on strategies such as low-cost index funds, diversification, and risk management.

In a nutshell, the EMH has had a significant impact on the way investors think about financial markets and has helped to shape investment strategies and regulatory policies.

25.8 APPLICABILITY OF THE EMH IN INDIA

In the Indian context, the applicability of the EMH has been a subject of much debate and research over the years. Here is a brief literature review on the same:

- Bhatia et al. (2017) conducted a study to test the weak form of the EMH in the Indian stock market using daily stock returns for the period 2006-2016. Their findings suggested that the Indian stock market is not weak-form efficient, as there is evidence of predictable patterns in stock returns that can be exploited for abnormal profits.
- Kumar and Kamaiah (2017) examined the semi-strong form of the EMH in the Indian stock market using event study methodology. They found that the Indian stock market is partially efficient in terms of reacting to news and events, but there is still room for abnormal profits to be made.
- Patro et al. (2009) analyzed the strong form of the EMH in the Indian stock market by examining insider trading activity. They found that insiders in the Indian market were able to generate abnormal profits, which suggests that the market is not strong-form efficient.
- Gupta and Sehgal (2013) investigated the efficiency of the Indian stock market in the context of the 2008 global financial crisis. They found that the market was largely efficient during the crisis period, but some anomalies suggest the presence of market inefficiencies.
- Bhattacharya and Mukherjee (2017) analyzed the efficiency of the Indian stock market during the period 2006-2016 using a combination of technical and fundamental analysis. Their findings suggested that the market is largely efficient, but there are still some anomalies that can be exploited for abnormal profits.

25.9 LET US SUM UP

The Efficient Markets Hypothesis is a financial theory that states that in an efficient market, a stock cannot perform better than the market as a whole. Three forms of efficiency can be found in the market: a) weak form, b) semi-strong form, and c) strong form. Weak form means that all past information has been absorbed by the market to arrive at the current prices, semi-strong form means that all publically available information is incorporated in the prices, and strong form means that all information (including inside information) is reflected by the market. The Capital Asset Pricing Model (CAPM) is an asset pricing model which predicts the expected return on a security. According to the model, the expected return on a security is linearly

and positively related to the systematic risk. The CAPM and the EMH are closely related. The CAPM assumes that the EMH holds good and the EMH can be tested using the CAPM. There are various methods to do this, but the Black, Jensen, Scholes (1972) is one of the most popular techniques.

25.10 KEY WORDS

The Efficient Markets Hypothesis (EMH) : The Efficient Markets Hypothesis (EMH) is a theory in finance that suggests that financial markets are “efficient,” which means that asset prices fully reflect all available information at any given time.

Weak Form of EMH : The weak form of EMH asserts that all historical price and volume data is already incorporated into current stock prices, making it impossible to profit from analyzing past trends or patterns.

Semi-Strong Form of EMH : The semi-strong form of EMH asserts that all **publicly available information** is already reflected in stock prices, making it impossible to achieve abnormal returns through fundamental analysis.

Strong Form of EMH : The strong form of EMH asserts that **all information, both public and private**, is already reflected in stock prices, making it impossible to achieve abnormal returns through any form of analysis or insider information.

Capital Asset Pricing Model (CAPM) : The CAPM defines that the expected return of an asset (or a security portfolio or stock) is linearly and positively related to its systematic risk (or undiversifiable risk).

Characteristic Line : A characteristic line is a straight line formed using regression analysis that summarizes a particular security’s systematic risk and rate of return.

25.11 SOME USEFUL BOOKS

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25.12 ANSWERS / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Suggests that financial markets are “efficient,” which means that asset prices fully reflect all available information at any given time. The hypothesis states that it is impossible to consistently achieve returns over average market returns because any new information that enters the market is immediately and fully reflected in the asset prices.
- 2) Weak Form, Semi-Strong Form, Strong Form
- 3) Rationality, Information is freely available, No asymmetricity of information, prices adjust quickly to new information, no individual investor can influence prices, no transaction costs.

Check Your Progress 2

- 1) The CAPM assumes that markets are efficient. The CAPM states that the expected return of an asset (or a security portfolio or stock) is linearly and positively related to its systematic risk (the part of total risk which is beyond the control of the firm or the investor; it is determined by various economic, social and political factors).
- 2) Refer to Section 25.5

UNIT 26 FINANCIAL STABILITY AND RELATED ISSUES

Structure

- 26.0 Objectives
- 26.1 Introduction
- 26.2 Concept of Financial Stability
- 26.3 Factors Affecting Financial Stability
- 26.4 Issues in Financial Stability
- 26.5 Challenges in Financial Stability
- 26.6 Risks and Financial Instability
- 26.7 Stability Measures for Ensuring Financial Stability
- 26.8 Financial Stability and Development Council
- 26.9 Financial Stability Report
- 26.10 Let Us Sum Up
- 26.11 Key Words
- 26.12 Some Useful Books
- 26.13 Answers / Hints to Check Your Progress Exercises

26.0 OBJECTIVES

After studying this unit, you will be able to:

- explain the concept of financial stability;
- identify factors affecting financial stability;
- discuss various issues in financial stability;
- state various challenges in financial stability;
- identify the major sources of financial instability; and
- suggest policy measures to attain financial stability.

26.1 INTRODUCTION

Financial stability is one of the important objectives of the Reserve Bank of India (RBI) besides macroeconomic stability. The RBI takes the key responsibility for promoting financial stability in the country and ensuring a sound financial system.

After the global financial crisis of 2008, India witnessed a deceleration in economic growth, a decline in exports, and volatility in the stock market. The

banking system too faced certain challenges such as growing NPAs. After the global financial crisis, the RBI started focusing rigorously on the financial stability of the Indian economy.

A sound financial system is the backbone of any country and so is the financial stability. In today's open economy, it is very important to have a sound financial system that can minimize the domino effect of any crisis. RBI has implemented several measures to promote financial stability during the time of crisis like strengthening the regulatory framework, improving risk management practices among financial institutions, and enhancing the supervisory and monitoring framework etc.

Several institutions and mechanisms have also been established by the RBI to promote financial stability. For instance, the Financial Stability and Development Council (FSDC) was established in 2010 to promote coordination and cooperation among the various financial regulators and other stakeholders in India.

Reserve Bank of India keeps a regular check on the financial health of the various financial components like banks, NBFCs, etc. to identify any kind of stress or risk to take timely action to avoid any adversities in the financial system and to attain the objective of financial stability and thereby economic growth.

26.2 CONCEPT OF FINANCIAL STABILITY

Financial stability refers to a financial system that is resilient to any shock or disruptions. The financial system consists of financial institutions like banks, non-banking financial institutions, financial markets, and financial instruments. Financial stability is achieved when the market volatility is low. The financial stability in a country helps ensure the efficient functioning of the entire financial system which would result in promoting economic growth and reducing poverty. Financial stability in a country ensures:

- a) sufficient availability of capital to financial institutions to ensure capital adequacy norms are met.
- b) sufficient risk management system to ensure protection against credit risk, liquidity risk, market risk etc.
- c) availability of well-developed infrastructure for trading and payment settlement.
- d) robust regulatory framework to ensure that all legal compliance is being observed and that laws and regulations are not overlooked.

26.3 FACTORS AFFECTING FINANCIAL STABILITY

Various factors have a bearing on the financial stability of the financial system of India and these factors are:

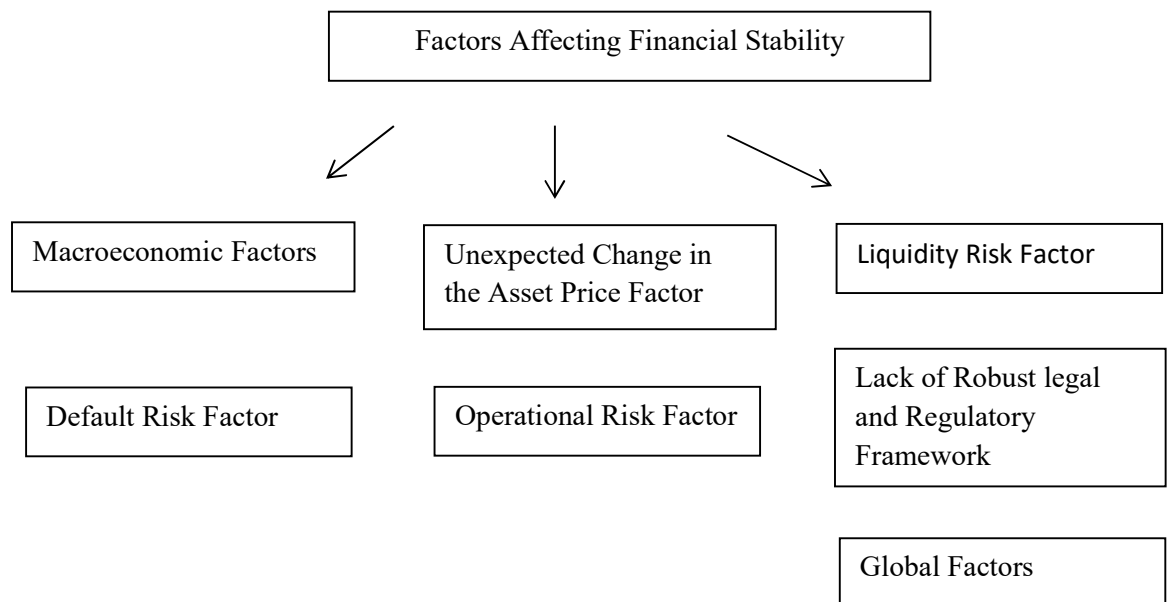


Fig. 26.1: Factors Affecting Financial Stability

- a) **Macroeconomic factors:** Factors like economic growth, interest rates, inflation, exchange rates, monetary and fiscal policy etc. have an impact on the financial stability of India. For instance, if inflation continues to grow and interest rates continue to increase, it may reduce the availability of credit to various sectors, result in high default rates, etc. and ultimately may lead to financial instability in a country.
- b) **Default risk factor:** The risk of default on loans and other credit instruments can have a significant bearing on financial stability. For instance, in case a large number of borrowers start defaulting on their loans, this may result in a liquidity crisis in the banks and will reduce the confidence of the savings holders, investors and others in the financial system of the country.
- c) **Unexpected change in asset prices factor:** The change in asset prices, like stock prices or real estate prices, have a bearing on the financial stability of the country. The rapid increase in asset prices may result in asset bubbles, which may burst at any time and can lead to financial instability in the country.
- d) **Operational risk factor:** Operational risk refers to the risk due to operational inefficiency which decreases the operational profits, sometimes results in losses and threatens the financial capabilities of any component of the financial system to meet the operational expenses or routine expenses and finally can impact the financial stability.
- e) **Liquidity risk factor:** Liquidity risk refers to the inability to meet short-term obligations in time. For instance, suppose at any point in time if the number of withdrawals from a bank suddenly increases beyond expectations then the bank may face liquidity issues which in turn increases the liquidity risk and may result in financial instability.

- f) Lack of robust legal and regulatory framework: The legal and regulatory framework have an impact on the financial stability of the various financial institutions. Lack of a robust legal and regulatory framework can lead to high risk in the financial system, and impose a threat on the financial stability as it fails to mitigate the impact of any kind of risk like financial crisis risk, global risk, liquidity risk etc.
- g) **Global factors:** Global factors like international trade relations and agreements, global economic trends, geopolitical events, etc. can significantly impact the financial stability of the financial system of a country. For instance, a global recession or a trade war between major economies can impact the financial stability of a country, like the war between Russia and Ukraine has an impact on the financial stability of the respective countries and their major trading partners.

We can conclude that financial stability is affected by many factors and if a country has a robust framework for regulating the financial system the impact of the above discussed factors may be reduced to a greater extent.

Check Your Progress 1

- 1) Define Financial Stability.

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- 2) Bring out the factors that affect the financial stability of an economy.

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- 3) Give an example of any global financial crisis that has an impact on the financial stability of India.

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26.4 ISSUES IN FINANCIAL STABILITY

There could be several issues that can arise in maintaining financial stability. These issues are:

- a) **Systemic risk:** Systemic risk refers to the risk that imposes a barrier in the efficient financial system that can have a severe impact on the entire economy of a country. There can be various sources of systemic risk like interconnections and interdependence between the financial institutions and markets, and the contagion effects that begin from the failure of one institution and engulf the entire institutions or markets resulting in widespread financial crisis. Systemic risk impacts financial stability in many ways as it can lead to the collapse of financial institutions and markets, can also result in a loss of confidence in the financial system and can also lead to a contagion effect.
- b) **Shadow banking:** Shadow banking refers to a system where the provision of credit and related financial services are done by entities that are not regulated, unlike banks. In the absence of a robust regulatory framework, such entities engage themselves in risky lending practices which increases the risk of default and can lead to systemic risk. There are various risks associated with shadow banking like it may build up systemic risk as it is not completely regulated and therefore it lacks accountability and may create difficulties in the financial system. Shadow banking can also lead to financial instability in the absence of any review mechanism.
- c) **Cyber-security:** Technology offers both benefits and limitations to any system as is the case with the financial system. Cyber-security threats like cyber attacks can pose a significant risk to the financial stability of any economy. Financial institutions like banks, non-banking financial institutions, investment firms, and payment systems, are the prime target for cybercriminals. Cyber-security threats can adversely impact financial stability in various ways like disrupting the flow of funds between financial institutions, through illegal access of sensitive financial information and using such information to engage in fraudulent activities like money laundering, unauthorized transactions etc. All this ultimately results in losing the confidence of the public in the financial system which can further result in a financial crisis. Therefore, it becomes very important to have a robust system of cyber-laws to deal with such kinds of cyber-attacks.
- d) **International spill-ins and spill-outs:** The international interconnections of financial markets can lead to international spillovers. Financial instability in one country has an impact on the other country which means that any financial disruption or financial shocks of one country can easily and quickly spread and affect the financial stability of the other country. All this may may result in a loss of confidence in financial markets and institutions, currency depreciation, asset price volatility etc. This could also result in increasing the threat of systemic risks besides exposing the economy to other vulnerabilities.

- e) **Too-big-to-fail:** This means that some financial institutions are so large that their failure could have a catastrophic impact on the economy. This perception of too-big-to-fail can result in moral hazard as these financial institutions are guaranteed by the government or RBI and therefore, they take excessive risks since they know that they will not be allowed to fail. Such institutions also erode the confidence of the general public as the public believes that these institutions are above the law and that their money (taxes) is being used to protect these institutions against failure.

Financial stability can only be maintained through a comprehensive and proactive approach of various parties like policymakers, regulators, and market participants. These participants must take the above issues in mind while planning for financial stability.

26.5 CHALLENGES IN FINANCIAL STABILITY

Planning financial stability is not an easy task for policymakers, regulatory bodies or market participants and several challenges exist in maintaining financial stability. These challenges are:

- a) **Highly Complex:** The financial system of India is increasingly becoming complex and has a wide range of financial instruments and institutions. Because of this complexity, the financial system is exposed to various types of risks like liquidity risk, credit risk, systemic risk etc. These risks impose a threat to the financial stability of the financial system of our country.
- b) **Rising Prices:** India has witnessed high inflation which can be a major threat to its financial stability. Inflation beyond a controllable limit can hurt the economy as it reduces the purchasing power of individuals, reduces the value of savings, and increases the cost of borrowing, which in turn discourages investments, and production and ultimately hurts the economy.
- c) **Fiscal deficit:** Fiscal deficit arises when the spending by the government exceeds its earnings. Such a situation leads to public debt and puts pressure on the government to increase taxes and cut spending. All this leads to a negative impact on the economy.
- d) **Growing Non-performing Assets (NPAs):** NPAs may be referred to as bad loans or loans that have not been repaid. Several banks in India are suffering from the problem of growing NPAs which may result in a liquidity crisis and can be a major threat to the financial stability as such bad loans reduce the profitability of financial institutions.
- e) **Organizational Deficiencies:** Several structural challenges also exist in the country that impose challenges on financial stability. These organizational deficiencies arise due to the existence of a large informal

sector, weak infrastructure, and the absence of a robust regulatory framework. Such factors make it difficult for the central bank and the government to effectively regulate and guard economic activities, which can be a threat to financial stability.

- f) **Outside influences:** Several external factors also make India vulnerable to managing financial stability. These outside influences can be changes in global commodity prices, fluctuations in exchange rates, shifts in global economic conditions etc. Such factors can have a bearing on a country’s trade and investment flows, which can have a ripple effect on the economy.
- g) **Unequal access to information:** This refers to a situation where one party in a transaction has more information than the other party. This leads to adverse selection and moral hazard, where the party with less information ends up making a poor decision involving excessive risk. Such a situation is challenging and leads to distrust or lack of confidence in the parties having less information and may lead to financial instability.
- h) **Behavioural biases:** Behavioural biases like herd behaviour and overconfidence end up in an investor making a decision that involves excessive risk and contributes to financial instability. These biases are difficult to predict and rationalize, as they are deeply rooted in individual and group psychology.

The objective of financial stability can be achieved through a comprehensive and coordinated approach that requires a deep understanding of the various components of the financial system and the economy in general and by identifying the various types of risks in particular and developing a system of robust regulatory framework.

Check Your Progress 2

- 1) Explain the major issues in financial stability.
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- 2) Discuss the various challenges related to financial stability.
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3) How a country like India can achieve financial stability?

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26.6 RISKS AND FINANCIAL INSTABILITY

Various types of risks that may trigger financial instability are:

- a) **Cyber-security threats** can adversely impact financial stability in various ways like disrupting the flow of funds between financial institutions, through illegal access to sensitive financial information and using such information to engage in fraudulent activities like money laundering, unauthorized transactions etc. All this ultimately results in losing the confidence of the public in the financial system which can further result in a financial crisis. Therefore, it becomes very important to have a robust system of cyber-laws to deal with such kinds of cyber-attacks.
- b) **Systemic risk** refers to the risk that imposes a barrier in the efficient financial system that can have a severe impact on the entire economy of a country. There can be various sources of systemic risk like interconnections and interdependence between the financial institutions and markets, and the contagion effects that begin from the failure of one institution and engulf the entire institutions or markets resulting in widespread financial crisis. Systemic risk impacts financial stability in many ways as it can lead to the collapse of financial institutions and markets, can also result in a loss of confidence in the financial system and can also lead to a contagion effect.
- c) **Credit risk:** Credit risk arises when the borrower fails to meet its financial obligations in time, or it may be defined as when a borrower defaults on a loan. To control such kind of risk various techniques are employed like checking the credibility score of the prospective borrower before lending credit or asking for collateral etc. By applying such kind of techniques, the risk of default may be minimized and financial stability may be ensured.
- d) **Market risk:** It refers to the risk that arises due to volatility in the financial market. Market rest triggers the risk of loss arising from changes in market price like asset price bubbles. To reduce the impact of the market risk and ensure financial stability in a country, the Reserve Bank of India and the various regulatory bodies apply tools like setting interest rates, regulating the activities of the Financial Institutions and implementing the macro-prudential policy.
- e) **Operational risk:** It may arise due to the lack of a strong internal system of an institution due to which problems like fraud, errors or

technical failures may trigger. To minimize the impact of operational risk various tools like risk management framework, internal control system and regulatory supervision techniques are adopted.

- f) **Liquidity risk:** It is a risk that arises from a shortage of liquid funds that are required to meet the financial obligation as and when they arise to manage the liquidity risk and ensure financial stability the Institutions apply various methods like maintaining adequate capital and liquidity reserves, implementing liquidity risk management etc.
- g) **Interest rate risk:** It refers to the risk that arises due to changes in the interest rates which have a bearing on the value of financial instruments and have an impact on the profitability of the financial institution. Interest rate risk can be minimized by applying various tools like stress testing, interest rates, risk management framework etc.
- h) **Political risk:** It refers to the risk that arises due to political events or actions like changes in government policy, geopolitical tensions, etc. Political risk hurts financial stability as it results in the loss of investors' confidence and can lead to regulatory changes to minimize political unrest. The various tools that may be applied are diversifying the geographic exposure, implementing a proper risk management framework etc.

26.7 STABILITY MEASURES FOR ENSURING FINANCIAL STABILITY

Various ways can help in achieving the objective of financial stability:

- a) **Robust regulatory framework:** A robust regulatory framework is an essential requirement for ensuring financial stability. A country requires regulations that promote fairness and transparency, accountability and risk management leading to a sound financial system.
- b) **Macro-prudential policies:** Macro-prudential policies help prevent systemic risk thereby resulting in financial stability. These policies include measures such as counter-cyclical capital buffers, reserve requirements, loan-to-value ratios etc.
- c) **Supervision and monitoring:** Through regular supervision and monitoring of financial institutions and keeping a check on the financial health of the institutions financial stability in the system can be promoted. Supervision and monitoring help in identifying whether the institutions are complying with the rules and regulations or not.
- d) **Crisis management framework:** It refers to a framework that consists of a set of policies and procedures that an institution may apply to manage and respond to a crisis. A crisis management framework is essential to ensure that financial crises are managed effectively.

- e) **Global cooperation:** Financial stability is a global issue, and therefore global cooperation is essential to ensure the stability of the financial system. Global cooperation includes cooperation between all global regulators, central banks of all countries and other global organizations to coordinate policy formulation and share information on different financial aspects.

Financial stability ensures the smooth functioning of the entire financial system which is very important for the growth of the company.

Check Your Progress 3

- 1) What are the various measures that may be adopted to ensure stability in the financial system?

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- 2) Explain the various types of risks to the financial stability of a country and how the same can be controlled.

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26.8 FINANCIAL STABILITY AND DEVELOPMENT COUNCIL

Established in December 2010 by the Government, the Financial Stability and Development Council (FSDC) serves as the apex forum to strengthen mechanisms for upholding financial stability, reinforcing inter-regulatory coordination, and fostering the development of the financial sector. Chaired by the Union Finance Minister, its esteemed members comprise the Governor of the Reserve Bank of India, the Finance Secretary and/or Secretary of the Department of Economic Affairs, the Secretary of the Department of Financial Services, the Chief Economic Adviser of the Ministry of Finance, the Chairperson of the Securities and Exchange Board of India (SEBI), the Chairperson of the Insurance Regulatory and Development Authority (IRDA), and the Chairperson of the Pension Fund Regulatory and Development Authority (PFRDA).

The Council's purview encompasses the macro-prudential supervision of the economy, including the oversight of large financial conglomerates, and the resolution of inter-regulatory coordination and financial sector development

matters. It also prioritizes initiatives concerning financial literacy and inclusion.

Furthermore, an FSDC Sub-committee, chaired by the Governor of the RBI, convenes more frequently than the full Council. All members of the FSDC also hold membership in the Sub-committee, alongside the four Deputy Governors of the RBI and the Additional Secretary of the DEA responsible for FSDC affairs. The Member Secretary of the Sub-committee is the Executive Director of the RBI, overseeing financial stability, with the Financial Stability Unit (FSU) of the RBI serving as the Secretariat for the Sub-committee.

26.9 FINANCIAL STABILITY REPORT

The Financial Stability Report (FSR) is a biannual report issued by the Reserve Bank of India (RBI). This report presents the consolidated evaluation of the Sub-Committee of the Financial Stability and Development Council (FSDC) regarding the risks to financial stability and the robustness of the financial system.

Here are the two latest Financial Stability Reports as released on 28th December 2023 and 28th June 2023 (source: rbi.org.in)

Highlights of Report released in December 2023

The global economy confronts a multitude of challenges, including the prospect of decelerating growth, substantial public debt burdens, escalating economic fragmentation, and persistent geopolitical tensions. However, the Indian economy and its domestic financial system demonstrate resilience, underpinned by robust macroeconomic fundamentals, sound financial institution balance sheets, moderating inflation trends, an improving external trade position, and ongoing fiscal consolidation efforts.

As of September 2023, scheduled commercial banks (SCBs) reported a Capital Risk-Weighted Assets Ratio (CRAR) of 16.8% and a Common Equity Tier 1 (CET1) ratio of 13.7%. Additionally, SCBs witnessed a decline in their gross non-performing assets (GNPA) ratio to a multi-year low of 3.2%, with the net non-performing assets (NNPA) ratio reaching 0.8%.

Macro stress tests for credit risk indicate that SCBs are anticipated to maintain compliance with minimum capital requirements. System-level CRAR projections for September 2024 stand at 14.8%, 13.5%, and 12.2%, respectively, under baseline, medium, and severe stress scenarios.

Furthermore, the non-banking financial companies (NBFCs) sector exhibits enhanced resilience, with a CRAR of 27.6%, a GNPA ratio of 4.6%, and a return on assets (ROA) of 2.9%, as recorded in September 2023.

Highlights of Report released in June 2023

Amidst heightened uncertainty globally due to banking system vulnerabilities in certain nations, ongoing geopolitical tensions, and moderating but elevated inflation, the global economy faces challenges.

Nevertheless, the Indian economy and its domestic financial system exhibit resilience, supported by robust macroeconomic fundamentals. Sustained growth momentum, decreasing inflation, a narrower current account deficit, escalating foreign exchange reserves, ongoing fiscal consolidation, and a sturdy financial system collectively propel the economy towards sustained growth.

Healthy balance sheets of both banks and corporations are fostering a new cycle of credit and investment, enhancing the outlook for the Indian economy.

As of March 2023, scheduled commercial banks (SCBs) achieved historical highs in their Capital Risk-Weighted Assets Ratio (CRAR) at 17.1% and Common Equity Tier 1 (CET1) ratio at 13.9%. Additionally, SCBs witnessed a decline in their gross non-performing assets (GNPA) ratio to a 10-year low of 3.9%, with the net non-performing assets (NNPA) ratio decreasing to 1.0%.

Macro stress tests indicate the ability of SCBs to meet minimum capital requirements even under severe stress scenarios. Projections for the system-level CRAR in March 2024, under baseline, medium, and severe stress scenarios, stand at 16.1%, 14.7%, and 13.3%, respectively.

Highlights of Report released in June 2024

The Reserve Bank released the 29th issue of the Financial Stability Report (FSR) on June 27, 2024, which reflects the collective assessment of the Sub-Committee of the Financial Stability and Development Council (FSDC) on the resilience of the Indian financial system and risks to financial stability.

- The global economy is facing heightened risks from prolonged geopolitical tensions, elevated public debt, and the slow progress in the last mile of disinflation. Despite these challenges, the global financial system has remained resilient, and financial conditions stable.
- The Indian economy and the financial system remain robust and resilient, anchored by macroeconomic and financial stability. With improved balance sheets, banks and financial institutions are supporting economic activity through sustained credit expansion.
- The capital to risk-weighted assets ratio (CRAR) and the common equity tier 1 (CET1) ratio of scheduled commercial banks (SCBs) stood at 16.8 per cent and 13.9 per cent, respectively, at end-March 2024.
- Scheduled Commercial Banks (SCBs)' gross non-performing assets (GNPA) ratio fell to a multi-year low of 2.8 per cent and the net non-performing assets (NNPA) ratio to 0.6 per cent at end-March 2024.

- Macro stress tests for credit risk reveal that SCBs would be able to comply with minimum capital requirements, with the system-level CRAR in March 2025 projected at 16.1 per cent, 14.4 per cent and 13.0 per cent, respectively, under baseline, medium and severe stress scenarios. These scenarios are stringent conservative assessments under hypothetical shocks and the results should not be interpreted as forecasts.
- Non-banking financial companies (NBFCs) remain healthy, with CRAR at 26.6 per cent, GNPA ratio at 4.0 per cent and return on assets (RoA) at 3.3 per cent, respectively, at end-March 2024.

Check Your Progress 4

- 1) What is FSDC? Why is it formed?
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- 2) Mention the highlights of the Financial Stability Report published in December 2023.
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26.10 LET US SUM UP

This unit discussed about financial stability which is one of the important objectives of the Reserve Bank of India (RBI). A robust financial system forms the backbone of any nation, and thus, maintaining financial stability is extremely essential. The Reserve Bank of India closely monitors the financial health of key components such as banks and NBFCs, identifying potential risks or stress points. This vigilance allows for timely interventions to prevent any disruptions to the financial system, thereby safeguarding financial stability and fostering sustained economic growth. Different factors affecting financial stability include macroeconomic factors, default risk factor, unexpected change in asset prices factor, operational risk factor, liquidity risk factor and lack of robust legal and regulatory framework. There could be several issues that can arise in maintaining financial stability. Some of them are systemic risk, shadow banking, cyber-security, international spill-ins and spill-outs etc. The objective of financial stability can be achieved through a comprehensive and coordinated approach that requires a deep understanding of the various components of the financial system and the economy in general and by identifying the various types of risks in particular and developing a system of robust regulatory framework.

26.11 KEY WORDS

Credit Risk	: It is the probability of a financial loss resulting from a borrower's failure to repay a loan.
Liquidity Risk	: Liquidity risk is the risk that a company or an individual will not have enough cash to meet its financial obligations (pay its debts) on time.
Foreign Exchange Risk	: Foreign exchange rate risk is the risk of financial impact due to exchange rate fluctuations.
Market Risk	: Market risk is the possibility that an individual or other entity will experience losses due to factors that affect the overall performance of investments in the financial markets.
Financial Stability	: Financial stability is defined as "a condition in which the financial system is not unstable".
Systemic Risk	: Systematic risk refers to the risk inherent to the entire market or market segment due to a certain event. Examples include natural disasters, changes in inflation rate, war, etc.
NBFCs	: Non-banking financial companies (NBFCs), also known as non-banking financial institutions (NBFIs), are financial institutions that offer various banking services but do not have a banking license.
Micro-prudential norms	: Micro-prudential regulation focuses on the stability of individual banking institutions and the protection of their customers rather than the economy as a whole.
Operational Risk	: Operational risk summarizes the uncertainties and hazards a company faces when it attempts to do its day-to-day business activities within a given field or industry.

26.12 SOME USEFUL BOOKS

Ramlall, I (2018), *Understanding Financial Stability*, Emerald Publishing

Moenjak, Thammark (2014), *Central Banking: Theory and Practice in Sustaining Monetary and Financial Stability*, Wiley Publication, 1st Edition.

26.13 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Financial stability refers to a financial system that is resilient to any shock or disruptions. The financial system consists of financial

institutions like banks, non-banking financial institutions, financial markets, and financial instruments.

- 2) Macroeconomic factors, default risk factors, unexpected changes in asset prices factor, operational risk factors, liquidity risk factors, lack of robust legal and regulatory framework, and global factors.
- 3) War between Russia and Ukraine

Check Your Progress 2

- 1) Systemic risk, shadow banking, cyber-security, international spill-ins and spill-outs, too-big-to-fail.
- 2) Highly Complex, rising prices, fiscal deficit, growing non-performing assets, organizational deficiencies, outside influences, Unequal access to information, and Behavioural biases.
- 3) The objective of financial stability can be achieved through a comprehensive and coordinated approach that requires a deep understanding of the various components of the financial system and the economy in general and by identifying the various types of risks in particular and developing a system of robust regulatory framework.

Check Your Progress 3

- 1) Robust regulatory framework, macro-prudential policies, supervision and monitoring, crisis management framework, global cooperation.
- 2) Cyber-security threats, systemic risks, credit risks, market risks, operational risks, liquidity risks, Interest rate risks, Political risk

Check Your Progress 4

- 1) Established in December 2010 by the Government, the Financial Stability and Development Council (FSDC) serves as the apex forum to strengthen mechanisms for upholding financial stability, reinforcing inter-regulatory coordination, and fostering the development of the financial sector.
- 2) Refer to Section 26.9

UNIT 27 NON-PERFORMING ASSETS (NPAs)

Structure

- 27.0 Objectives
- 27.1 Introduction
- 27.2 Profile of Non-Performing Assets in India
 - 27.2.1 Nature of Non-Performing Assets
 - 27.2.2 Magnitude and Trend of NPAs
 - 27.2.3 Major Causes of NPAs
- 27.3 Approach of RBI Towards Non-Performing Assets
- 27.4 Impact of Non-Performing Assets
- 27.5 Measures to Tackle the Problem of NPAs
- 27.6 Effectiveness of Action Taken to Curb NPAs
- 27.7 Recent Policy Measures towards NPAs
- 27.8 Let Us Sum Up
- 27.9 Key Words
- 27.10 Some Useful Books and References
- 27.11 Answer/Hints to Check Your Progress Exercises
- 27.12 Terminal Questions

27.0 OBJECTIVES

By going through this unit, you will be able to:

- explain the concept of non-performing assets;
- identify the different kinds of NPAs;
- indicate the magnitude of NPAs in Indian Banks;
- identifying major causes of NPAs;
- describe the impact of NPAs on various sectors of the economy;
- highlight the role of RBI in NPAs;
- analyse the measures taken to tackle the problem of NPAs; and
- suggest measures for having effective policies for solving the menace of NPAs.

27.1 INTRODUCTION

The banking industry is a lending institution that helps millions of people get sufficient funds to carry out their operations. The banks are responsible for building financial stability in the nation. Over the world, NPAs in the banking sector are growing. NPAs are found to be at the centre of the economic problem of the banks.

NPAs refer to the classification used by financial institutions for loans and advances that are in default or arrears. A loan is in arrears when the principal is due, or interest payments are late, or missed. NPAs can be classified as substandard assets, doubtful assets, or loss assets depending on the length of time overdue and the probability of repayment. It is important for both the borrower and the lender to be aware of performing versus non-performing assets.

Banks improve the allocation of resources by lending money to priority sectors of the economy, but non-performing loans assume critical importance since they reflect the asset quality, credit risk and efficiency in the allocation of resources to productive sectors. The level of NPAs creates serious concern for the stability of the banking system. The financial health of the economy is deteriorating due to an upsurge in credit defaulters in the country.

NPAs can be manageable. It depends on its magnitude and its past due period. In the short term, banks can tolerate a good amount of NPAs. A healthy financial sector is necessary for the growth of the economy. A profitable banking sector also has more capacity to absorb shocks and provide relative stability to the economy. The increasing trend in NPAs is testing banks' strength and resilience (Guar & Mohapatra, 2020). The accumulation of NPAs in India cannot be attributed to a single event nor can it be confined to a particular timeline. The growth rate in the NPA level shows that the problem is evident not only with small-sized banks but also with major banks.

In this unit, we focus on various aspects of NPAs such as nature, growth and magnitude of NPAs, factors responsible for NPAs, and impact of NPAs on the economy. Policy measures taken by the authorities have also been analysed along with suggestions for prevention of NPAs.

27.2 PROFILE OF NON-PERFORMING ASSETS IN INDIA

The NPAs refer to a class of loans or advances that are in default or arrears. For measurement purposes NPAs are classified into the following two types:

- i) *Gross NPAs*: This indicates the absolute amount of the total value of loans that are due within 90 days in a particular quarter or financial year.
- ii) *Net NPAs*: This indicates the net value of NPAs after the bank makes specific provisions for it. It is arrived at by subtracting the doubtful and unpaid assets from the gross NPAs.

The above distinction reflects the quality of loans made by banks. Gross NPAs Ratio = Gross NPAs/Gross Advances. Net NPAs are those types of NPAs in which the banks deduct the provisions regarding NPAs. It shows the actual burden of banks.

$$\text{Net NPAs} = \text{Gross NPAs} - \text{Gross Advances} - \text{Provisions}$$

Gross NPAs is an absolute amount which is considered irrecoverable and shows the total value of NPAs for the bank in a particular financial year. On the other hand, Net NPAs provide the exact value of NPAs after making specific provisions for the particular asset. NPAs are listed on the balance sheet of a bank or lender. A loan can be classified as an NPA at any point during the term of the loan or at its maturity. Banks usually provide a grace period before classifying an asset as non-performing. Until and unless the loan asset is generating regular income with the expected usual commercial risk, that asset is known as a performing asset. If it fails to generate the expected revenue, then it is identified as an NPA.

We must distinguish that all the performing assets of a bank are known as *standard assets*. Such standard assets are not treated as NPAs because such assets have a distinctive risk profile and do not require any supplementary provisions.

27.2.1 Nature of Non-Performing Assets

A loan is said to be in arrears when the principal amount or interest payments are late or missed and a loan is in default when the lender considers the loan agreement to be broken and the debtor is unable to meet the obligations. You should note that NPAs are recorded on a bank's balance sheet only after a prolonged period of non-repayment by the borrower.

Considering the nature of NPAs, the RBI has categorized them into three broad categories depending on the period of default. Banks are required to classify NPAs into one of three categories according to how long the asset has been nonperforming: sub-standard assets, doubtful assets, and loss assets. Let us define these categories.

- i) **Sub-Standard Asset:** Any loan or advance is a sub-standard asset if it remains outstanding or NPA for a period less than or equal to 12 months.
- ii) **Doubtful Asset:** Any loan or advance is a doubtful asset if it remains outstanding or NPA for more than 12 months and is already classified as a sub-standard asset.
- iii) **Loss Asset:** Any loan or advance is a loss asset if it remains *uncollectible* or has little to no recovery value, as its continuance as a bankable asset is un-suggestible. Therefore, the bank may consider some recovery value left in it as the asset has not been written off wholly or in parts. Given this situation, banks are forced to accept that the loan will never be

repaid, and must record a loss on their balance sheet. The entire amount of the loan must be written off completely.

Although the most common NPAs are term loans, there are other forms of NPAs as well (Naveen J., 2022):

- i) Overdraft and cash credit (OD/CC) accounts left out-of-order for more than 90 days.
- ii) Agricultural loans/advances that are approved for short-period crops may also be treated as NPA if the instalment of principal or interest thereon remains unpaid for two crop seasons. On the other hand, if a loan is approved for long-period crops, it may be treated as NPA, if the instalment of principal or interest thereon remains unpaid for one crop season.
- iii) Expected payment on any other type of account is overdue for more than 90 days.

A loan becomes a non-performing asset after 90 days of non-repayment of loans. After non-payment, the bank asks the borrower to liquidate any assets or mortgage as a part of the loan agreement. If the borrower does not pledge any assets, then the bank may write off the loan as bad debt and sell it to a 'bad bank' (for example, National Asset Reconstruction Company Limited (NARCL)). The bad banks aim to ease the burden on banks by taking the bad loans off their balance sheet. The NARCL is a public-sector asset reconstruction company in India. It is established to acquire and aggregate NPAs.

27.2.2 Magnitude and Trend of NPAs

It was only in the mid-1990s; post the economic liberalization when the country entered into the new millennium that the banks realized the sudden compounding of bad loans. The NPA situation essentially propelled in the mid-2000s following over-optimism in the economy specifically between 2006 and 2008, on the back of strong economic growth and pending infrastructure projects completed on time and within specified budgets, a thing largely unheard of till then (Palli, 2020).

The fact that the Gross NPA ratio (defined as the ratio of Gross NPA to total assets) of scheduled commercial banks was more than 11 per cent in March 2018 (more than 14 per cent in the case of public sector banks). Over time, however, this ratio has declined. It was placed at a much better 8.2 per cent in March 2020 and 7.3 per cent in March 2021. In March 2024 the Gross NPA ratio was 2.8 per cent, according to the Financial Stability Report (FSR) of the RBI.

The occurrence of fraud as a proportion of the gross advances of PSBs dropped from a peak of 1.32 per cent during the financial year 2013-14 to 0.05 per cent during the financial year 2021-22. The improved detection and

reporting accompanied by the comprehensive steps taken to check frauds have resulted in a decline in the occurrence of such frauds (ET Bureau, 2022).

Credit is required for faster growth of the economy but any kind of macroeconomic shock can impact those outstanding advances badly and may turn them non-performing. According to the RBI, total net NPAs as of March 31, 2022 amounts to ₹204,225.90 crore. The share of public sector banks in these NPAs stood at 75.75 per cent. The rest pertains to the private sector and foreign and small finance banks. See Table 27.1.

Table 27.1: Net NPAs in Indian Banks as of March 31, 2022

(Amount in ₹ Crores)

Banks	Net NPAs	Percentage
Public Sector Banks	154745.39	75.75
Private Sector Banks	43733.24	21.44
Foreign Banks	3022.72	1.48
Small Finance Banks	2724.55	1.33
Total	204225.90	100.00

Source: RBI's Statistical Tables Relating to Banks in India 2022.

Movement of non-performing assets (NPAs) of scheduled commercial banks

<https://dbie.rbi.org.in/BOE/OpenDocument/1608101727/OpenDocument/opendoc/openDocument.faces?logonSuccessful=true&shareId=2>

Due to increasing loan failures, the share of non-performing advances has increased substantially. The pace of credit defaulters has been rising due to a lack of proper monitoring and verification. The big corporate individuals make use of flaws in the law and fly away with the money of banks. There was a huge increase in NPAs in the fiscal year 2017-18 after that there was a decrease in NPAs (Jain *et al.*, n.d.).

It has been observed that a major portion of NPAs is contributed by several top industrialists — a dirty dozen in the case of India. Generally, the NPAs in agriculture and the priority sector are comparatively lower than that of the corporate houses (Sikdar, 2020). Most of the current NPAs originated during the mid-2000s, during a period of high economic growth and positive business outlook. This led to an easy flow of credit. The impact of ownership (public and private sector banks) significantly influences the gross and net NPAs. The NPAs of public sector banks were higher than that of private sector banks.

After the global financial crisis of 2008, the Indian economy stagnated and so did the repayment capacity of these businesses. This contributed to what is

now known as India's Twin Balance Sheet (TBS) problem, where both the banking sector (that gives loans) and the corporate sector (that takes and has to repay these loans) have come under financial stress. The twin balance sheet problem refers to a situation where the borrower (corporate sector) is under stress due to high debt, and the lender (banks) is under stress due to non-repayment of loans (NPAs). The balance sheets of both the borrower and the lender have problems.

Most banks follow Early Warning Systems (EWS) for recognition of probable NPAs, but the actual procedure followed varies from bank to bank. Under this system, the banks monitor and track the portfolios of their customers regularly. The EWS helps the banks to check the asset quality of the borrowers (corporate firms). The EWS (software) helps the banks to identify signs of distress which is not possible to check manually.

As regards the decline in NPAs Chandrasekhar (2022) has rightly commented that "this may have occurred partly through the use of write-offs that erode the capital base of banks and also because of the time-bound moratorium on debt repayments announced as part of measures to address the effects of the pandemic on small units and other selected borrowers. In such circumstances, even though lending to micro, small and medium enterprises was partly guaranteed by the government, a rise in the NPAs ratio and further erosion of bank capital seems inevitable."

27.2.3 Major Causes of NPAs

Accumulation of NPAs is a recurring attribute of financial crisis and financial stress as NPAs typically grow when a credit boom turns into a bust. Several factors affect the profitability of the banks. The factors can broadly be categorised into bank-specific and macroeconomic factors (Das & Uppal, 2021).

- i) The bank-specific factors are:
 - a) non-performing advances,
 - b) deposits,
 - c) non-interest income,
 - d) interest income,
 - e) operational efficiency, and
 - f) capital adequacy.
- ii) The macroeconomic factors are:
 - a) GDP growth,
 - b) rate of inflation, and
 - c) interest rate.

The following are major reasons for the emergence and continuance of NPAs in India (Naveen J., 2022):

- i) From 2000-2008, the Indian economy was in a boom phase and banks, especially public sector banks, started lending extensively to companies.
- ii) With the financial crisis in 2008-09, corporate profits decreased and the Government banned mining projects. The situation became serious with the substantial delay in environmental permits, affecting the infrastructure sector – power, iron, and steel.
- iii) Taken together all these factors resulted in a rise in prices of raw materials and a shortage of supply.
- iv) Moreover, the relaxed lending norms adopted by banks and foregoing analysis of the financials and credit ratings, especially of the big corporate houses.
- v) Due to the absence of:
 - a) credit information bureau,
 - b) poor lending strategies and low scrutiny by banks,
 - c) wilful defaults,
 - d) economic slowdown and government policies,
 - e) inadequate risk and supervision,
 - f) lack of adequate capital,
 - g) low level of technology,
 - h) dishonest and poor credit documentation,
 - i) concentration in few customers/groups,
 - j) corruption, political and government interventions etc.

There have been frauds of high magnitude that have contributed to rising NPAs. The top management of the banks was forced by politicians and bureaucrats to throw good money after bad in the case of unscrupulous borrowers. (Selvarajan, & Vadivalagan, 2013). Several bad loans were advanced by the banks often by compromising due diligence.

In short, the reasons for growth in NPAs range from bank-specific to government policies. The factors that affect NPAs are kinds of advances, the credit value of borrowers, collateral securities, the absence of legitimate observation, inadequacy and inaccuracy in the credit assessment standards that have been set, the death of potential earning members in the family and the absenteeism of entrepreneurial information and sense on the borrower's part (Singh & Arora, n.d.).

It is important to remember that business units engaged in infrastructural activities, base metal, textile, food processing, chemical, engineering etc. failed to make repayment of interest and principal amount on their loans and advances. Negligence and ineffectiveness of recovery tribunals etc. The NPA levels for some banks crossed the 20 per cent level. This meant that banks reached negative capital levels. In other words, the true value of bank deposits was below par.

Check Your Progress 1

- 1) Explain the nature of NPAs in India.

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- 2) How are NPAs recorded in Banks' Balance sheets?

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- 3) Highlight the magnitude and trend of NPAs in recent years.

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- 4) What are Early Warning Systems (EWS) for the recognition of probable NPAs?

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- 5) Explain the concept of a Twin Balance Sheet (TBS).

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27.3 APPROACH OF RBI TOWARDS NON-PERFORMING ASSETS

The RBI took a simple but holistic approach to empower the banks. With the help of the Securities Exchange Board of India (SEBI), in 2015 the RBI introduced the 'Strategic Debt Restructuring Scheme'. The objective of the scheme was to help financial institutions in recovering loans by taking over

distressed companies. This gave the banks substantial power in the management of companies, which defaulted in repaying the borrowings (Palli, 2020).

For tackling the problem of NPAs, the RBI in 2019 issued fresh guidelines. According to these guidelines, defaults are to be recognized within 30 days. During this 30-day period, lenders may decide on the resolution strategy, including the nature of the resolution plan and its implementation. All lenders must put in place board-approved policies for the resolution of stressed assets, and it is expected that the lenders initiate the process of implementing a resolution plan even before a default. Further, the lenders shall report credit information on all borrowings aggregating over ₹ 50 million and submit a weekly report on any default. If a resolution plan is to be implemented, the lenders shall enter into an inter-creditor agreement (Palli, 2020).

The RBI has come up with stringent norms so that the growth of NPAs can be put under control. The Insolvency and Bankruptcy Code of 2016 plays an important role in the recovery of assets of those creditors whose case has been filed with the National Company Law Tribunal (NCLT). Only time will tell how successful has the RBI been in controlling the NPA growth in the sector (Agarwala & Agarwala, 2019).

The RBI has underlined regulatory changes in India in recent years. These range from a prudential asset-liability management framework for domestic banks and curbs on total leverage to the systematic plugging of regulatory arbitrage between banks and other financial entities. Recently, RBI Governor cautioned banks against any build-up of asset-liability mismatches, saying both are detrimental to financial stability and hinted that the ongoing crisis in the US banking system seems to have emanated from such mismatches. He emphasised the fact that the ongoing US banking crisis also clearly shows the risks of private cryptocurrencies to the financial system (The Hindu Business Line, March 17, 2023).

The RBI has been taking a series of measures to strengthen Indian banks and the financial system as a whole. Depositors should repose faith in its ability to safeguard their interests. But in the long term, it is the full implementation of international standards in regulation, supervision, governance, resolution and deposit insurance that will best help convince the markets of the stability of the banking system.

27.4 IMPACT OF NON-PERFORMING ASSETS

A strong banking sector is important for the Indian economy. The failure of the banking system may hurt other sectors. The rising NPAs are a cause of concern. It on the one hand reduces their interest earning and, on the other hand, also affects their future deposits and increases their operating cost as the cost of recovery of NPAs will go up. NPAs generate a vicious cycle of effects on the sustainability and growth of the banking system, and if not managed properly could lead to bank failure (LSS, 2014). For the bank or

lender, interest earned on loans acts as a major source of income. NPAs place a financial burden on the lender. A large volume and significant number of NPAs over some time is an indication of the fact that the bank is in jeopardy. This situation makes RBI, the regulator, aware of the weakening financial fitness of the bank.

NPAs hurt the rate of profit of the Indian banks. NPAs negatively affect banks' ability to generate income and profitability. As such, it is incumbent on banks to keep constant track of their NPAs to avoid adverse effects on their liquidity and capacity to grow. If the volume of NPAs continues to increase over some time, it can threaten the financial health of the bank.

Higher NPAs influence the price of loans and interest rates, which in turn affects the confidence of investors, lenders, and depositors equally. While higher interest rates will directly impact the investors in need of loans for the creation of infrastructural and industrial projects, they also cause poor recovery of funds, which will affect the credit creation and revenue stream of the banks. The non-recovery of loans affects credit creation and further affects the financial soundness of an economy (Hafsal et al, 2020).

There is a positive relation in the banking industry between total income and gross NPAs, whereas a negative relation is in the case of total profit and gross NPAs. This means that with an increase in total income, there is an increase in NPAs on the contrary there will be a decrease in profit with an increase of NPAs (Jain et al n.d.). NPAs have a significant negative impact on the profitability of scheduled commercial banks measured in terms of return on assets and return on equity.

It enhances the risk of credit default and even affects the image of banks in the nation. Lender's reputation is decreasing due to the continuous increase in the defaulters, not paying back the bank's money on time. The overall reputation of the banks is at stake due to increased NPAs and the upsurge in the number of depraved loans (Jain, et al, n.d.)

The rise in NPAs occurred with the deterioration of the balance sheets of non-financial firms, and this twin balance sheet crisis contributed significantly to the deceleration of growth in the late 2010s. NPA write-offs do erode the capital base, which has been only partially neutralised in the case of public sector banks by the government's recapitalisation effort (Chandrasekhar, 2022).

NPAs increase borrowing costs for the general public, because banks may increase their lending interest rates to maintain balance in their costs and profits. A rise in NPAs beyond the risk threshold can have significant effects on the stability of the banks, reducing consumer confidence in the functioning of these banks. In general, NPAs affect the morale and confidence of all the stakeholders associated with the operation of banks. The challenge of NPAs has made the banks risk-averse. They hesitate to extend

loans to business ventures that may remotely appear risky for fear of aggravating an already high level of NPAs.

Check Your Progress 2

- 1) Examine the steps taken by RBI about NPAs.

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- 2) In what ways NPAs affect the growth of the Indian economy?

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- 3) What is the relationship between NPAs and Banks' profitability?

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- 4) Highlight the impact of rising NPAs on Banks' capacity to create credit.

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27.5 MEASURES TO TACKLE THE PROBLEM OF NPAS

Until the mid-eighties, management of NPAs was left to the banks and the auditors. In 1991, the Narasimhan Committee on the Financial System suggested that banks should classify their advances into four broad groups, viz. (i) standard assets; (ii) substandard assets; (iii) doubtful assets; and (iv) loss assets. Following this, prudential norms relating to income recognition, asset classification and provisioning were introduced in 1992 in a phased manner.

In 1998, the Narasimhan Committee on Banking Sector Reforms recommended a further tightening of prudential standards to strengthen the prevailing norms and bring them on par with evolving international best practices. With the introduction of 90-day norms for the classification of NPAs in 2001, the NPA guidelines were brought to par with international standards (LSS, 2014).

NPAs can be manageable, but it depends on how many there are and how far they are past due. In the short term, most banks can take on a fair amount of

NPAs. However, if the volume of NPAs continues to build over some time, it threatens the financial health and future success of the lender.

Generally, banks keep NPAs in their books in the expectation of recovering the money or making provisions for it. The banks or the lenders have options to recover their losses, including taking possession of any collateral. If they are unable to recover or sell off the loan at a discount to a collection agency they write it off entirely as bad debt.

The decline of NPAs is essential to improve the profitability of banks and fulfil the capital adequacy norms as per the Basel Accord (Rajput et al, 2011). Central Repository of Information on Large Credits (CRILC) was set up to share data for loans over ₹50 million with all banks. The CRILC data included the status of each loan; whether it was performing or had become NPA or was heading in that direction. This system helped the banks identify warning signs and helped segregate the habitual offenders (Palli, 2020).

To empower the lenders, a new legislation was enacted in the form of the Insolvency and Bankruptcy Code (IBC) in 2016. More than 50 per cent of NPA matters are being resolved through the IBC. It laid down separate insolvency procedures for companies, individuals and partnership companies giving a maximum period of 180 days for resolution extendable by 90 days. The NCLT was set up under the IBC, 2016 and at the pre-admission stage 4452 cases were disposed of which helped creditors obtain dues of ₹2 trillion; and various restructured post-liquidation or adjudication (Palli, 2020).

The National Asset Reconstruction Company (NARCL) along with the India Debt Resolution Company (IDRCL) will take over the first set of bad loans from banks and try to resolve them. In addition to the statutory provisions already available to lenders for recovery and resolution, the other measures are:

- i) Recovery of Debts and Bankruptcy Act, 1993,
- ii) Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest Act, 2002, and
- iii) Insolvency & Bankruptcy Code, 2016 (IBC).

The government launched '*Mission Indradhanush*' in 2015 to make the working of public sector banks more transparent and professional to curb the menace of NPAs in future. Banks continue to pursue recovery actions initiated in written-off accounts through various recovery mechanisms available.

An important amendment was made in 2021 in the SARFAESI Act, 2002, which gives not only the right of ownership to the banks on such properties but also provides the right of control and management of such forfeiture-secured assets. Now, banks and financial institutions may receive guidance and assistance from Asset Reconstruction Companies (ARCs) to collect their NPAs.

As a precautionary measure, banks set aside a proportion of their profits to meet the risk created by the NPAs. This amount is known as provisioning for non-performing assets. As such, banks keep aside a certain amount from their profits in a particular quarter for NPAs. This step is taken as a precautionary measure in anticipation that assets can turn into losses in the future. Such provisions are made by Banks based on different NPA categories. This provision also depends upon the type of bank. For instance, Tier-I and Tier-II banks have different provisioning norms. By this method, banks can maintain a healthy book of accounts by provisioning for bad assets.

There is a need to ensure that the banking system recognizes financial distress early, takes prompt steps to resolve it, and ensures fair recovery for lenders and investors so that the banking sector functions without stress.

27.6 EFFECTIVENESS OF ACTION TAKEN TO CURB NPAS

The state of the banking system is reflected in the banking stability indicator. BIS gives an overall assessment of changes in underlying conditions and risk factors that have a bearing on the stability of the banking sector. Despite the efforts of the RBI and recommendations of Basel committees, the problem of an upsurge in NPAs is continuing (Jain, et al, n.d.)

Banks need to take precautions regarding credit assessments and should take measures in pre- and post-sanction of the loans to avoid slippages and standard assets of NPAs. The procedures for loans and securities hypothecated need to be checked. NPA and their provisions, interest calculations, repayment schedule, and audit of bank borrowers need to be critically analyzed. There should be a system of periodic review of credit card holders' accounts (Jain, et al, n.d.). Timely action is essential to ensure the future growth of the Bank.

The prudential and provisioning norms and other initiatives taken by the regulatory bodies have pressurized banks to improve their performance and consequently resulted in the trim down of NPAs as well as improvement in the financial health of the Indian banking system. The government in August 2019, announced a merger of 10 public sector banks into 4 banks reducing the public sector banks to 12 from 27. This will help improve the profitability of public sector banks and consequently reduce total gross NPAs. Remarkably, rigorous methods have been and shall continue to be adopted to address the situation surrounding NPAs. Instituting the use of third-party data sources in PSBs for comprehensive due diligence across data sources at the sanction stage itself, to mitigate risk on account of misrepresentation and fraud. Despite, RBI's efforts to control NPAs, there is room for further action in the following way (Palli, 2020):

- i) Records of borrowers should be available to banks to form an opinion and conduct financial background checks of prospective borrowers.

- ii) The credit team of banks should also undergo periodic training to update their skills and industry benchmarks along with credit evaluation sheets, which should be upgraded regularly.
- iii) Data analytics can also be a significant tool to conduct effective periodic reviews and audits.

The banks are supposed to focus their actions on reducing their NPAs and operating costs to improve their profitability (Das & Uppal, 2021). The banks should follow proper procedures for the dispatch of loans. There is a need to take time-oriented action to hinder the evolution of NPAs in India.

27.7 RECENT POLICY MEASURES TOWARDS NPAS

The issue of NPAs is of considerable significance for the regulators as the intensification of the same has always been the precursor to the banking crisis and banking failures worldwide. One must recognize the centrality of NPAs in ensuring the stability of scheduled commercial banks. The proactive approach to make provisions on the probable losses even for the standard assets is a globally accepted prudent norm set by the IFRS 9 standard. The RBI also highlighted the importance of keeping the expected loss-based forward-looking provisions for the safety of commercial banks in India.

Tracking the significant increase in credit risk is a novel addition to the new accounting norm since it will enable the banks to heighten their diligence to protect themselves from the risks of bad loans. The ECL methodology takes into account the historical probability of default trends as well as current and future economic scenarios and predictions. Thus, it significantly changes banks' incentives by inclining them to manage and dispose of bad loans much more actively than the existing process. The ECL method is expected to bring prudence in lending since banks will be able to better monitor their loan slippage and track rating movements (Bandyopadhyay, 2023).

Customising systemic controls to domestic banking, the RBI rightly urged banks to make extra provisions against exposure to the top 20 business houses that face higher external risks. The nudge by the RBI is more towards building a buffer, banks can decide on how much they would want to set aside if they have exposure to any of the 20 conglomerates. The prudential provisioning will reflect as part of the provisioning buffer held by banks. (The Hindu Business Line, March 27, 2023).

The crux of the recent Supreme Court ruling is that it affirms the right of a borrower to be given a fair hearing before the NPA account is declared as fraud. On March 27, 2023, the Supreme Court in the case *State Bank of India vs Rajesh Agarwal* ruled that the principles of natural justice must be applied to borrowers, who should be given a hearing before banks classify loan accounts as fraudulent. The court also said the decision to classify a loan as fraudulent should be made by a reasoned order, given the potentially serious

civil consequences. The SC ruling may lead to fewer cases of fraud being registered but could also lead to more accountability and punishment in such cases. Given the situation, even a small improvement on this front would be a huge step (Mint, March 31, 2023).

Check Your Progress 3

- 1) Give an outline of measures taken by banks to curb the growth of NPAs.
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- 2) What are the suggestions made by the Narasimhan Committee in 1998?
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- 3) Analyse the effectiveness of measures for tackling the problem of NPAs.
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- 4) Why has RBI urged banks to make extra provisions against mismatch between assets and liabilities?
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- 5) In what way Central Repository of Information on Large Credits (CRILC) will help in reducing NPAs?
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27.8 LET US SUM UP

NPAs are loans or advances made by a financial institution, on which both principal and interest are unpaid for a specified period. NPAs are those loans that have ceased to generate income for the bank. They are recorded on a bank's balance sheet after a prolonged period of non-payment by the borrower. NPAs can be classified as substandard assets, doubtful assets, or loss assets, depending on the length of time overdue and the probability of repayment. The increase in non-performing advances hurts the rate of profit

Several factors like prevailing macroeconomic conditions, sectoral issues, global business environment, delayed recognition of stress by banks, aggressive lending during upturns, improper risk pricing and poor credit underwriting were attributed towards NPA build-up. One of the primary reasons for stressed assets can be attributed to lethargic due diligence and an ineffective sanctioning process. A large proportion of NPAs arose because of exogenous shock. Most of the NPAs arose due to defaults by private sector non-financial firms.

The government and RBI regularly issue guidelines and have taken several initiatives aimed at the resolution of long-standing stressed assets on the books of banks as well as timely identification and recognition of stress immediately upon default and taking corrective action for mitigation of the same. For the borrower, if the asset is non-performing and does not pay the interest, it can hurt their creditworthiness and growth. Also, it will hamper their ability to obtain a loan in future.

Banks faced a significant challenge of rising NPAs in 2011 which peaked at 11.18 per cent in the fiscal year 2018. Even the government is allowing the banks to make provision for NPAs. It is important that both the borrower and lender must be aware of the NPAs. RBI has introduced stringent policy norms and banks have to make their NPA numbers public from time to time.

27.9 KEY WORDS

- Bank for International Settlements (BIS)** : It is an international financial institution owned by central banks that “fosters international monetary and financial cooperation and serves as a bank for central banks.”
- ECL approach** : This approach requires the estimation of credit losses through the life of a financial instrument and subsequently the present value of such credit losses, as on the reporting date.
- Fraud** : This is indicative of deceitful acts. Bank fraud is a purposeful act of omission or conduct by any person in the course of a banking transaction or the bank’s books of accounts. Fraud denotes a false statement made knowingly.
- IFRS 9 standard** : IFRS 9 specifies how an entity should classify and measure financial assets, financial liabilities, and some contracts to buy or sell non-financial items.
- Twin Balance Sheet (TBS) problem** : This deals with two balance sheet problems. One with Indian companies and the other with Indian Banks. Thus, TBS is a two-fold problem for the Indian economy which deals with overleveraged

companies and bad-loan-encumbered banks. This can be explained by the fact that the corporations over-expand during a boom, leaving them with obligations that they can't repay. So, they default on their debts, leaving bank balance sheets impaired, as well. This combination then proves devastating for growth. This has been an issue lingering for decades in the Indian economy, albeit in different forms and structural names.

Willful defaulter : According to RBI, a willful defaulter is any unit which has defaulted in repaying the lender. Even when it could be repaid. Willful default occurs when a debtor willfully defaults on the repayment of a loan.

Zombie lending : Zombie lending is the extension of new credit or prolonging of existing loans to low-productivity firms. While such policies may stabilise the economy in the short run, they risk transforming transitory shocks into phases of delayed recovery and permanent productivity and output losses. It refers to extending credit to entities that cannot repay.

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27.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Considering the nature of NPAs, the RBI has categorized them into three broad categories depending on the period of default. These categories are Sub-Standard Asset, Doubtful Asset, and Loss Asset.
- 2) NPAs are recorded on a bank's balance sheet only after a prolonged period of non-repayment by the borrower. When the banks accept that the loan will never be repaid, they record a loss on their balance sheet. The entire amount of the loan must be written off completely.
- 3) Refer to Sub-section 27.2.2
- 4) Under this system, the banks monitor and track the portfolios of their customers regularly. The EWS helps the banks to check the asset quality of the borrowers (corporate firms). The EWS (software) helps the banks to identify signs of distress which is not possible to check manually.
- 5) The twin balance sheet problem refers to a situation where the borrower (corporate sector) is under stress due to high debt, and the lender (banks) is under stress due to non-repayment of loans (NPAs). The balance sheets of both the borrower and the lender have problems.

Check Your Progress 2

- 1) Defaults are to be recognized within 30 days, The Insolvency and Bankruptcy Code of 2016 plays an important role in the recovery of assets of those creditors whose case has been filed with the National Company Law Tribunal (NCLT). prudential asset-liability management

framework for domestic banks and curbs on total leverage to the systematic plugging of regulatory arbitrage between banks and other financial entities.

- 2) A strong banking sector is important for the Indian economy. The failure of the banking system may hurt other sectors. Refer to Section 27.6 for the detailed answer.
- 3) NPAs hurt the rate of profit of the Indian banks. NPAs negatively affect banks' ability to generate income and profitability. It on the one hand reduces their interest earning and, on the other hand, also affects their future deposits and increases their operating cost.
- 4) Higher NPAs influence the price of loans and interest rates, which in turn affects the confidence of investors, lenders, and depositors equally. While higher interest rates will directly impact the investors in need of loans for the creation of infrastructural and industrial projects, they also cause poor recovery of funds, which will affect the credit creation and revenue stream of the banks.

Check Your Progress 3

- 1) Refer to Section 27.9
- 2) In 1998, the Narasimhan Committee on Banking Sector Reforms recommended a further tightening of prudential standards to strengthen the prevailing norms and bring them on par with evolving international best practices. With the introduction of 90-day norms for the classification of NPAs in 2001, the NPA guidelines were brought to par with international standards.
- 3) NPAs can be manageable, but it depends on how many there are and how far they are past due. In the short term, most banks can take on a fair amount of NPAs. However, if the volume of NPAs continues to build over some time, it threatens the financial health and future success of the lender. Also, refer to section 27.8.
- 4) The nudge by the RBI is more towards building a buffer, banks can decide on how much they would want to set aside if they have exposure to any of the 20 conglomerates.
- 5) CRILC) was set up to share data for loans over ₹50 million with all banks. The CRILC data included the status of each loan; whether it was performing or had become NPA or was heading in that direction. This system helped the banks identify warning signs and helped segregate the habitual offenders.

27.12 TERMINAL EXERCISES

- 1) Do you agree that there is a declining trend in the stock of NPAs? Give reasons in support of your answer.

- 2) “NPAs in the banking sector are growing and are found to be at the centre of the economic problem of the banks in India”. Explain this statement and discuss the problem created by growing NPAs.
- 3) How do NPAs reflect the asset quality, credit risk and efficiency in the allocation of resources to productive sectors? Explain the impact of NPAs on the stability of the banking system in India.
- 4) Why does the high level of NPAs cause worry to any financial institution? What measures have been taken to regulate the problem of NPAs?
- 5) “Most of the NPAs originated during the mid-2000s, during the period of high economic growth and positive business outlook. This led to an easy flow of credit”. Give reasons for this rise in NPAs.
- 6) Do you agree that there is a need to have a proper set of policy measures for recognizing NPAs early? What prompt steps to resolve and ensure fair recovery for lenders would you recommend?



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UNIT 28 FOREIGN EXCHANGE STABILITY AND RELATED ISSUES

Structure

- 28.0 Objectives
- 28.1 Concept of Foreign Exchange Stability
- 28.2 Basic Concepts
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28.0 OBJECTIVES

After going through this unit, learners will be able to:

- explain the concept of foreign exchange stability;
- identify the key issues related to foreign exchange stability; and
- suggest measures to achieve foreign exchange stability.

28.1 CONCEPT OF FOREIGN EXCHANGE STABILITY

Foreign exchange stability refers to the consistency and predictability of exchange rates between different currencies. Alternatively, foreign exchange stability, also known as exchange rate stability, refers to a situation in which the currency of an economy maintains a relatively consistent or predictable value over other foreign currencies over some time. Exchange rate stability is a desirable goal for many countries and has several important implications for their economies and international trade. Foreign exchange stability is a key concern for governments, businesses, and investors, as it affects the cost of imports and exports, the competitiveness of industries, and the overall economic health of a country. Maintaining a stable exchange rate is often a delicate balancing act, requiring careful policy choices and international cooperation to mitigate risks and support economic growth.

Let us discuss certain basic concepts related to exchange rate stability.

28.2.1 Meaning of Exchange Rate

Many times, we hear through the mass media such as Zee Business, Bloomberg TV, CNN, CNBC, BBC or print media such as The Times of India, Business Line, etc. that the US dollar has appreciated against the Indian rupee or Indian rupee has become stronger against the US dollar, Sterling Pound, Japanese Yen, etc. What does it mean? The media frequently discusses the ups and downs in exchange rates. An exchange rate means the price of the currency of a country like the Indian rupee in terms of the currency of another country like the US dollar. So, when the media report that the exchange rate between India and the US is ₹ 83.17/\$1, then it means that one US dollar is equal to ₹ 83.17. In other words, it means that one US dollar can buy 83.17 Indian rupees. As the dollar appreciates against the Indian rupee, then one dollar can buy more Indian rupees or as the dollar depreciates against the Indian rupee, then one dollar can buy less Indian rupees. Media may report an exchange rate in direct and indirect quotes the meaning of which are given below.

A **direct quote** measures how many units of domestic currency can be bought by one unit of a foreign currency while an **Indirect quote** measures how many units of foreign currency can be bought by one unit of domestic currency. Thus, the direct quote between the Indian rupee (domestic currency) and the US dollar (foreign currency) can be expressed as follows:

$$\text{\$1} = \text{₹ 83.17}$$

The indirect quote between the Indian rupee (domestic currency) and the US dollar (foreign currency) can be expressed as follows:

$$\text{₹ 1} = \text{\$ 0.0120}$$

28.2.2 Exchange Rate Systems

- a) **Fixed Exchange Rate:** Under this regime, a country's currency is pegged to another currency or a fixed value like gold. This provides stability but can be challenging to maintain in the long run. Under this system, the exchange rate is determined by the central bank (e.g., RBI in India). The exchange rate so fixed is also called **the official exchange rate**. The central bank ensures that the exchange rate should not go beyond or fall below the exchange rate fixed. To understand how the fixed exchange rate system works, let us examine Fig. 28.1. We assume that the Reserve Bank of India (RBI) desires to keep the exchange rate between the US dollar (\$) and Indian rupee (₹) fixed at ₹ 50/\$. But the equilibrium exchange rate in the foreign exchange market the exchange rate is ₹ 60/\$ as determined by the intersection of demand curve D and

supply curve S_1 at point E_1 , then to bring the exchange rate to the desired level of ₹ 50/\$, the RBI will have to sell the dollars equal to Q_1Q_2 so that the supply can increase from S_1 to S_2 intersecting demand curve D at point E_2 . Thus, we have a new equilibrium point E_2 at which the exchange rate reaches to the desired level of ₹ 50/\$. Now the question is how does the central bank sell Q_1Q_2 dollars? The answer is the arbitration process. In the arbitrage process, there is a person, known as an arbitrageur, who can buy one dollar from the RBI by paying ₹ 50 and then sell the same dollar in the foreign exchange market for ₹ 60 to earn a profit of ₹ 10 per dollar. Thus, an arbitrageur can earn a total profit of ₹ 10 multiplied by Q_1Q_2 . One point is also relevant here i.e., on selling Q_1Q_2 dollars the foreign exchange reserves held by RBI fall. Thus, when a currency (US dollar in the present case) is overvalued, then foreign exchange reserves fall if a fixed exchange rate system is adopted. If the equilibrium exchange rate in the foreign exchange market falls below the desired level, then the RBI can buy the required amount of dollars to bring it above. It will increase the foreign exchange reserves.

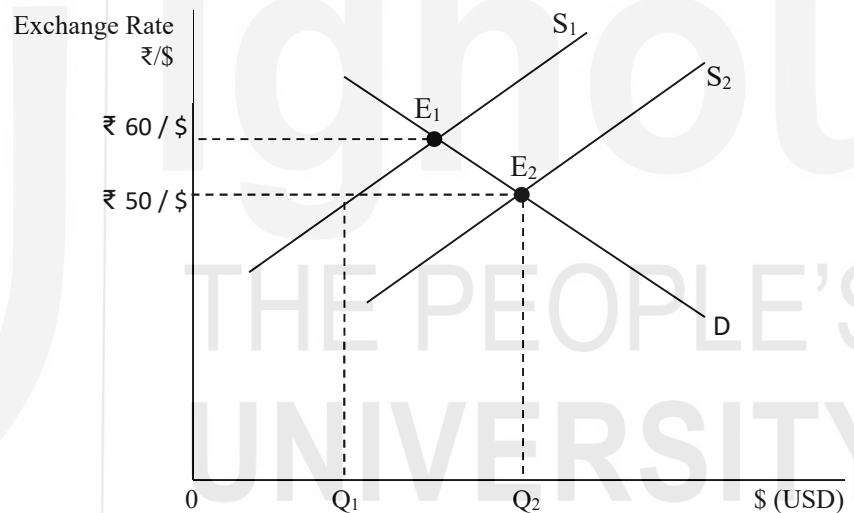


Fig. 28.1: Fixed Exchange Rate System

- b) **Floating Exchange Rate:** In this regime, the exchange rate is determined by market forces, such as supply and demand. This allows for flexibility but can lead to volatility. Under this system, the exchange rate is determined based on the demand and supply of the foreign currency concerned. Alternatively, market forces of demand and supply for the foreign currency decide the equilibrium exchange rate. To understand this, consider the US dollar as a foreign currency against the Indian rupee, then the exchange rate between the US and India is determined by the demand for and supply of the US dollar. When the demand for the US dollar is equal to its supply, the corresponding exchange rate is known as the equilibrium exchange rate. It is evident that shifts in supply and demand affect the exchange rate and cause it to rise or fall. Let us understand the system through Fig. 28.2. In Fig. 28.2, D_1 denotes the demand for the US dollar and S_1 does the supply of the same. Both the curves are intersecting at the point E_1 . Therefore, point

E_1 is the equilibrium point where the exchange rate is e_1 called the equilibrium exchange rate. Now, suppose that the supply of the US dollar increases causing the supply curve to shift S_2 . S_2 intersects D_1 at E_2 . Thus, the new equilibrium exchange rate is e_2 . Thus, we conclude that an increase in supply with constant demand leads fall in the exchange rate. Opposite will happen when the supply falls with constant demand. Now suppose that due to some reason, the demand for the US dollars increases to D_2 with constant supply at S_1 . As a result, a new equilibrium exchange rate e_3 is obtained. Therefore, an increase in demand with constant supply leads increase in the exchange rate. Opposite will happen if the demand falls with constant supply. However, demand and supply both can change simultaneously, accordingly exchange rate too fluctuates.

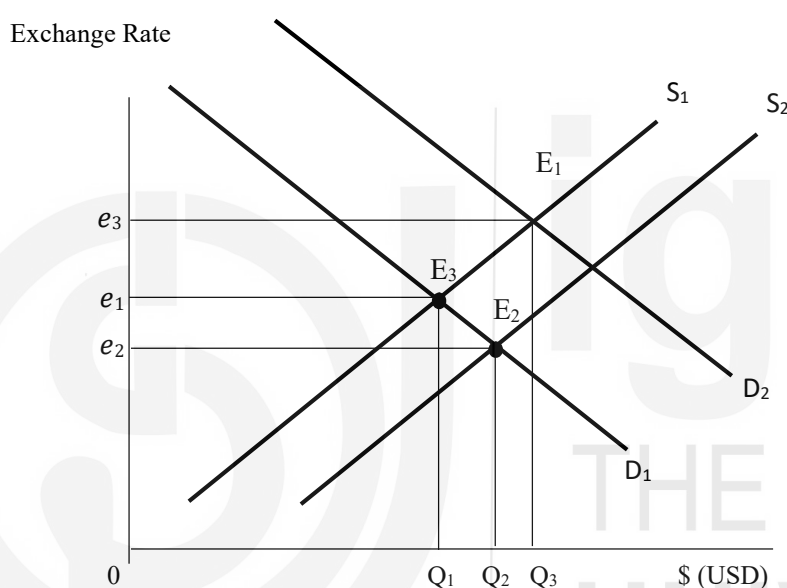


Fig. 28.2: Floating Exchange Rate System

28.2.3 Factors Affecting Demand for Foreign Exchange

Some of the factors that affect demand for foreign exchange are discussed as follows:

1) To pay import bills

To import goods and services from abroad our country needs foreign currency. So, if an Indian trader wants to import cars from the US, then he will demand US dollars to pay for the cars. Similarly, if Air India wants to buy some new airbuses from the UK, then it will demand Sterling Pound to pay for the air buses. Thus, imports create demand for foreign currencies.

2) To repay international loans

Indian corporations or governments demand foreign currency to repay the loans taken from foreign countries. Thus, repayment of international loans creates demand for foreign exchange.

3) Arbitrage purpose

Suppose, there are two foreign exchange markets X and Y in which the exchange rates between the Indian rupee and the US dollar are ₹ 50 per dollar and ₹ 55 per dollar respectively. Then, a person who wants to gain from these exchange rate differences will buy the US dollars from X and sell the same in Y and will earn ₹ 5 per dollar as profit. This is known as the arbitrage process. This is obvious that the demand for the US dollar increases in market X.

4) Speculation

Speculation means the opportunity for profit due to the difference between the prices of a good/share/currency etc. at two points in time. So, if we expect that the US dollar will appreciate in future, it means the US dollar is relatively cheaper now, therefore, one may buy or demand US dollars today and sell the same in future at a higher price can earn a profit.

5) To give gifts, and grants to foreign countries

Indian residents or governments may decide to give gifts, grants etc. for some purposes. To do so they will demand foreign currency. For example, the Indian government may decide to gift 10 million US dollars; the government will demand 10 million US dollars.

6) Investment in foreign countries

Indian investors may decide to invest in foreign countries. For example, if TATA group decides to take over Jaguar (a car company in Great Britain), the latter will demand Sterling Pound for payment of the consideration. Similarly, if a resident of India wants to buy equity shares of a US company, then (s)he will demand US dollars to do so.

7) To visit foreign countries

Indian residents will demand Sterling Pound, US dollars, Australian dollars, Canadian dollars etc. if they want to visit Great Britain, the US, Australia, and Canada respectively. The purpose of the visit may be tourism, higher study, etc.

8) To remit the money

Foreigners may be working in India and they will earn in Indian currency. To remit these earnings to their parents, spouses, relatives etc. in their home countries, they will demand foreign currencies.

28.2.4 Factors Affecting Supply of Foreign Exchange**1) Receipts of exports**

Exports of goods and services are one the major sources of foreign exchange supply. So, if an Indian car company exports some cars to the US dollars will move to India.

2) Repayment of international loans

Indian corporations or governments might have decided to give loans to other countries, then their repayments will bring foreign exchange.

3) Arbitrage purpose

Suppose, there are two foreign exchange markets X and Y in which the exchange rates between the Indian rupee and the US dollar are ₹ 50 per dollar and ₹ 55 per dollar respectively. Then, a person who wants to gain from these exchange rate differences will buy the US dollars from X and sell the same in Y and will earn ₹ 5 per dollar as profit. This is known as the **arbitrage** process. It is obvious that the supply of dollars increases in market Y.

4) Speculation

Speculation means the opportunity for profit due to the difference between the prices of goods/ stocks/ currency, etc. at two points in time. So, if we expect that the US Dollar will depreciate in future, it means the US Dollar is relatively expensive now, therefore, one may sell US Dollars today to escape from future loss.

5) Gifts and grants from foreign countries

Foreign residents or governments may decide to give gifts, grants etc. to India, for some purposes. As a result, foreign exchange will move to India.

6) Investments from foreign countries

Foreign investment in a domestic company can lead to an increase in the supply of foreign exchange in several ways. Foreign exchange refers to the currency of a foreign country, and it is used to facilitate international trade and investment. When foreign investors invest in a domestic company, they often need to acquire the domestic currency to make their investment.

7) Visiting the home country (i.e., India, in our Case)

Foreign residents or NRIs may decide to visit India for tourism, higher study, or other purposes. Therefore, they will need Indian currency to pay in India. Hence, they will exchange the currencies against the Indian rupees. In this way, the foreign exchange reserves of India will increase.

8) Foreign remittances

Indian residents or Indian corporations may be working or operating outside India and they will earn in foreign currencies. To remit these earnings to their parents, spouses, relatives etc. in India, they will

demand Indian currency in exchange for the foreign currencies available to them.

Check Your Progress 1

- 1) What is the need for foreign exchange stability in a country?

- 2) Briefly describe various factors affecting the Supply of Foreign Exchange.

- 3) Compare fixed exchange rate system with floating exchange rate system.

28.3 ISSUES IN FOREIGN EXCHANGE STABILITY

Here are some key aspects and related issues in foreign exchange stability:

1) Managed or Dirty Floating

A “managed float,” also referred to as a “dirty float,” is a regime for exchange rates where the exchange rate is neither fully set nor completely free. Instead, central bank intervention maintains the currency’s value within a range relative to another currency or a basket of currencies. Some countries are accused of manipulating their exchange rates to gain an unfair trade advantage. This can lead to international tensions and trade disputes. In today’s global economy, currency exchange rates play a crucial role in shaping a country’s economic performance. Nearly all economies actively manage their currencies, with central banks or governments frequently intervening to control their values. According to data from the International Monetary Fund (IMF) in 2014¹, 82 countries and regions, representing 43 per cent of economies, employed a managed float system, making it the most prevalent exchange rate regime.

¹<https://asia.nikkei.com/Markets/Currencies/IMF-finds-more-countries-adopting-managed-floating-exchange-rate-system>

2) Exchange Rate Policy

The exchange rate policy of an economy can be influenced by its economic goals. For example, a country may pursue a strong currency policy to control inflation or a weak currency policy to boost exports. For example,

.....in the 1990s, India's economy was experiencing hardship. 39% of the money the government was collecting at the time went toward paying interest. After cutting its fiscal deficit to 7.8% of GDP, India faced the risk of being branded a defaulter on the global stage. A devaluation of the Indian rupee was necessary in light of this problem. The process of devaluation involves a country lowering the value of its currency on the global market while maintaining its intrinsic value. Similar measures were taken by India to increase the cost of its imports and decrease the price of its exports.

Source: <https://www.bookmyforex.com/blog/1-usd-inr-1947-till-now/>

3) Parallel Foreign Exchange Systems

Parallel foreign exchange systems are found in many developing countries, where they combine a market-driven exchange rate that primarily applies to financial transactions, and sometimes a portion of trade transactions, alongside one or more fixed exchange rates. These arrangements are often implemented in response to a balance of payments crisis. In one scenario, governments establish a legal parallel or dual foreign exchange market for financial transactions. This move is aimed at mitigating the immediate impact of a depreciation in the exchange rate on domestic prices while retaining some level of control over capital outflows and international reserves. In another scenario, strict foreign exchange controls limit access to official markets, leading to the emergence of an illicit parallel market. As the authorities respond to a worsening balance of payments situation by tightening and expanding exchange controls, the illicit market gains prominence. Strong limitations that have led to external imbalances and the growth of a substantial parallel market where the currency currently trades at nearly a 60 per cent premium characterize Burundi's foreign exchange market. One of the authorities' main short-term goals going ahead is to stabilize the exchange rate and lessen these distortions². Informal trade (i.e., parallel foreign exchange market) in South Asia was 50 per cent of formal trade between 1993 and 2005³.

²www.elibrary.imf.org

³ <https://blogs.worldbank.org/endpovertyinsouthasia/capital-controls-and-parallel-exchange-rates-south-asia>

4) Political Instability

Political stability affects foreign exchange rates significantly and raises demand for a currency. Switzerland, for example, is renowned for its political stability, adherence to neutrality, and avoidance of international confrontations. Because of this, the Swiss franc is seen as a haven in foreign exchange trading and is frequently used to offset currency risk. Conversely, political unrest, coups, and protests destabilize the economy and devalue the local currency. As a result, there is less demand for and more risk associated with keeping the relevant domestic currency.

5) Trade Balances

Consider the distribution of national output as

$$Y = C^d + I^d + G^d + X$$

Where,

Y = National output

C^d = Consumption of domestically made goods and services

I^d = Investment in domestically made goods and services

G^d = Government expenditure on domestically made goods and services

X = Exports of domestically made goods and services

We know that in the world of globalization or the case of an open economy, people do not consume only domestically produced services, investment is not restricted to domestic goods and services only and similarly, government purchases are not limited to domestic goods and services only. Therefore, we can say that

$$C = C^d + C^F$$

$$I = I^d + I^F$$

$$G = G^d + G^F$$

where the super-script F on C, I and G show the consumption of, investment in and government expenditure on foreign goods and services respectively.

Now, the national income identity can be re-written as follows:

$$Y = C - C^F + I - I^F + G - G^F + X$$

$$Y = C + I + G + X - C^F - I^F - G^F$$

$$Y = C + I + G + X - (C^F + I^F + G^F)$$

Since $C^F + I^F + G^F$ means the total imports denoted by IM and therefore

$$Y = C + I + G + X - IM$$

The difference $X - IM$ is called net exports (NX) and many times called trade balance and therefore national income accounts identity in an open economy becomes

$$Y = C + I + G + NX$$

$$NX = Y - (C + I + G)$$

Thus,

$$\text{Net exports} = \text{National output} - \text{Domestic spending}$$

Now, Consider the national income account identity again

$$Y = C + I + G + NX$$

$$Y - C - G = I + NX$$

The left-hand side of the equation is called national saving (S) and therefore

$$S = I + NX$$

$$S - I = NX$$

Thus, the net exports are equal to the excess of savings over investments. The difference between savings and investment is called net capital outflow (or net foreign investment) because the difference shows the flow capital i.e., if S is greater than I, then the excess amount is invested in the rest of the world which causes the outflow of the capital from the domestic land and if S is less than I, then the shortage is met from the rest of the world which causes inflow of the capital. Thus, positive NX means net capital outflow which means there is trade surplus and negative NX means net capital inflow which means trade deficit. In this way, net exports are financed by the net capital outflow. Now, we consider a small open economy in case of which this is assumed that the domestic interest denoted by r^* is determined by the world interest rate denoted by r . Thus,

$$r^* = r$$

We know that the investment is a function of interest rate and therefore we have

$$I = I(r)$$

Further, since, S depends upon Y, C and G and Y is supposed to be constant because it is determined by the factors of production, G is

constant and C is a function of disposal income which is also determined by Y and therefore, the S becomes constant which is denoted by $\bar{S}$.

Hence,

$$\bar{S} - I(r^*) = NX$$

The last equation clearly shows that net exports or trade balance depends upon the world interest rate r^* and $\bar{S}$.

Real Interest rate

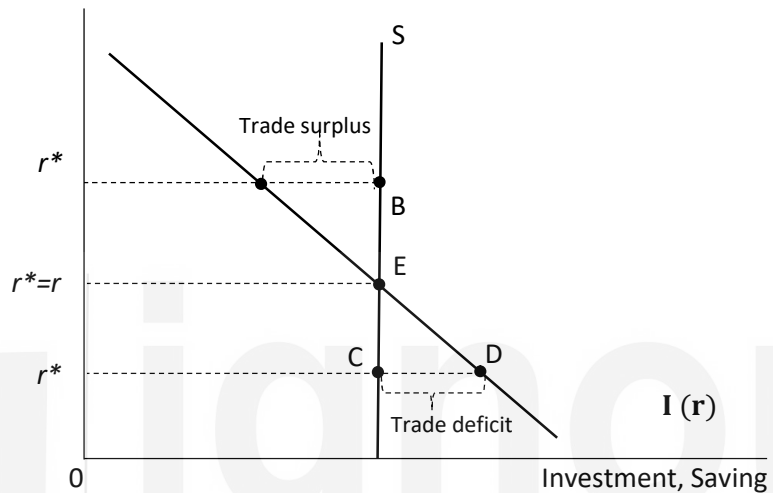


Fig. 28.3: Relationship between Real Interest Rate, Saving and Investment

Fig. 28.3 shows that with constant saving line and downward-sloping investment function determine the interest rate r if the economy is closed and it is in equilibrium at E . But when the economy is open and the world interest rate $r^* > r$, then there is a trade surplus equal to AB and when $r^* < r$, there is a trade deficit equal to CD . When $r^* = r$, then there is zero trade balance. Thus, in the case of a small open economy, the world interest rate determines the trade balance. Now, we see the relationship between the real exchange rate and the net export

We know that

Real Exchange Rate

$$= \text{Nominal Exchange Rate} \times \frac{\text{Price level in the domestic country}}{\text{price level in the foreign country}}$$

Let

$$\alpha = \text{Real exchange rate}$$

$$e = \text{Nominal exchange rate}$$

And

$$\theta = \frac{\text{Price level in the domestic country}}{\text{Price level in the foreign country}}$$

Therefore,

$$\alpha = e \times \theta$$

Now, for a given value of e , α will rise if θ rises or vice versa. Therefore, when the real exchange rate increases, then it means foreign goods have become relatively inexpensive and it will be beneficial to buy goods from abroad. As a result, imports from the home country will increase. At the same time, foreigners will not buy from the domestic country due to low relatively expensive goods as compared to their own country. As a result, the exports of the home country will decrease. In this way, the real exchange rate causes imports to increase and exports to fall. Overall, net exports fall. Hence, there is a negative relationship between the net exports and the real exchange rate. Mathematically, it can be said that net export is a negative function of the real exchange rate. Symbolically,

$$NX = f(\theta)$$

Graphically,

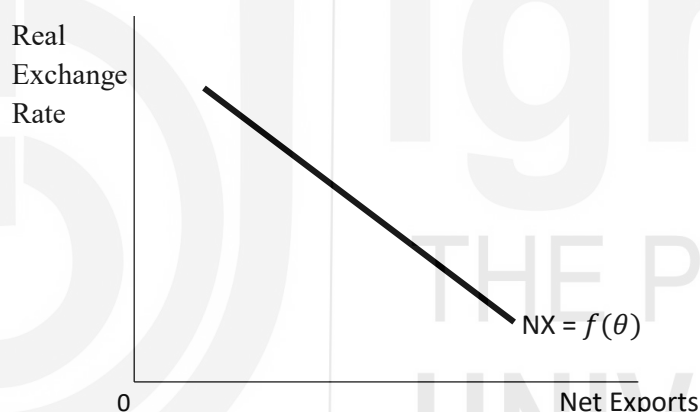


Fig. 28.4: Relationship between Real Interest Rate and Net Exports

Thus, the relative worth of currencies is shaped by the demand for them, which, in turn, is heavily influenced by trade dynamics. When a country consistently exports more than it imports, resulting in a trade surplus, there is a strong demand for its products and, consequently, its currency. According to the basic principles of supply and demand, high demand leads to higher prices and an increase in the value of the currency, which appreciates.

Conversely, when a country imports more than it exports, creating a trade deficit, the demand for its currency is relatively weaker. As a result, currency prices tend to decrease. In the context of currency, this means the currency depreciates or loses value.

6) Inflation and Interest Rates

Inflation and interest rates are closely intertwined, and their relationship can significantly affect exchange rates. Countries strive to strike a

delicate balance between these two factors, but managing this interplay is often intricate. Lower interest rates stimulate consumer spending and foster economic growth, typically exerting a positive impact on a country's currency value. However, if consumer spending surges to the extent that it outpaces supply, it can lead to inflation, which isn't necessarily a negative outcome. Nonetheless, lower interest rates typically do not attract foreign investment. On the other hand, higher interest rates tend to lure foreign investors, which, in turn, heightens the demand for a country's currency. Finding the right equilibrium in this regard can be challenging, and the effect on a country's exchange rate can be complex. In general, when inflation is high, it tends to weaken a currency as its purchasing power diminishes. This makes goods more expensive and can discourage investors. Conversely, when inflation is low, more capital flows into the country, bolstering the currency's value due to increased purchasing power and thus improving its exchange rate. In this regard, you should recall the IS-LM model in a closed economy and the **Mundell-Fleming Model** in an open economy.

7) Global Events

Events like financial crises, geopolitical tensions, and economic downturns in major economies can significantly impact exchange rates and disrupt stability. 2008 saw a severe decline in global currency values. The global financial crisis was a global issue that affected the Indian economy in a ripple fashion. After the Great Depression, this was the most serious crisis. Covid-19 caused an enormous crisis in 2020 that affected the entire planet. Every country's economy had difficulties as a result of the pandemic, including India. The value of the Indian rupee experienced fluctuation. The country did, however, recover more quickly than expected thanks to its fortitude and aggressive initiatives. Chortane and Pandey (2022)⁴ conducted an event study to find the impact of the Russia-Ukraine war on the value of global currencies against the US dollar. The study shows that the conflict between Russia and Ukraine affected the value of all major world currencies negatively, but a region-by-region analysis reveals that the currencies of the Pacific region significantly appreciated, the currencies of the Middle East and Africa (ME&A) are negligible, and European currencies (especially the Russian rouble, Czech koruna, and Polish zloty) depreciated against the USD. Further, the study shows how the currencies of Poland and the Czech Republic have depreciated considerably compared to the US dollar as a result of the financial and economic restrictions placed on Russia and their closeness to the conflict zone. Moreover, the Pan-American, European (especially the Polish zloty and the Russian rouble), and

⁴Chortane, S. G., & Pandey, D. K. (2022). Does the Russia-Ukraine war lead to currency asymmetries? A US dollar tale. *The Journal of Economic Asymmetries*, 26, Article e00265. <https://doi.org/10.1016/j.jeca.2022.e00265>

Middle Eastern and African (ME&A) currencies have all benefited greatly from the Russian Central Bank's announcement to tie the rouble with gold.

28.4 MEASURES TO MAINTAIN FOREIGN EXCHANGE STABILITY

Here are certain ways to maintain foreign exchange stability in an economy.

1) Central Bank Intervention

Central banks can use their foreign exchange reserves to buy or sell their currency in the foreign exchange market to influence its value. By doing so, they can mitigate excessive currency fluctuations and maintain stability. This is often done to counteract extreme currency appreciation or depreciation.

2) Fixed Exchange Rate Systems

Some countries opt for fixed or pegged exchange rate systems where the value of their currency is fixed or pegged to another stable currency, such as the U.S. dollar or the euro. Maintaining this peg requires active central bank intervention and foreign reserves to ensure that the exchange rate remains stable. However, the fixed exchange rate system has its limitations.

3) Managed Float

In a managed float system, exchange rates are allowed to fluctuate within a certain range, but central banks will intervene when the rate moves outside of that range to prevent extreme volatility.

4) Fiscal and Monetary Policies

Governments can use fiscal and monetary policies to influence exchange rates indirectly. For example, interest rate adjustments can affect the attractiveness of a currency to foreign investors. Tightening or loosening monetary policy can influence exchange rates.

5) Economic Reforms

Structural reforms in the economy, such as improving infrastructure, reducing trade barriers, and addressing fiscal deficits, can boost confidence in the currency and help maintain its stability.

6) Exchange Rate Bands

Implementing an exchange rate band system can allow for some degree of fluctuation while preventing extreme movements. The central bank intervenes when the exchange rate reaches the upper or lower bounds of the band.

7) **Transparency and Communication**

Central banks and governments should communicate their exchange rate policies clearly to the public and financial markets. Transparency can help manage expectations and reduce uncertainty.

8) **Foreign Exchange Reserves**

Accumulating and maintaining foreign exchange reserves provides a financial buffer that can be used to stabilize the currency during periods of volatility. However, building and maintaining sufficient reserves can be expensive.

9) **International Cooperation**

Collaboration with other countries and international organizations can help stabilize exchange rates. This can include currency swap agreements, coordinated interventions, and harmonized policies.

You should note that the approach to maintaining foreign exchange stability can vary from one country to another based on their economic conditions, policy objectives, and the global economic environment. Additionally, exchange rate stability is not always the primary goal; some countries may prioritize other economic objectives, such as inflation control or economic growth, which can affect exchange rates.

Check Your Progress 2

- 1) Explain the relationship between Real interest rate, Saving and Investment.
.....
.....
.....
- 2) Describe the relationship between real interest rate and net exports.
.....
.....
.....
- 3) State some of the latest issues of foreign exchange stability.
.....
.....
.....
- 4) What are your suggestions for maintaining foreign exchange stability?

28.5 LET US SUM UP

Foreign exchange stability refers to the ability of a country to maintain a relatively stable and predictable exchange rate for its currency in the foreign exchange market. In other words, it means that the value of a country's currency remains relatively constant over time, which can be important for economic and financial stability. Foreign exchange stability is essential for a country's economic well-being and international trade. Achieving and maintaining this stability involves a complex interplay of economic policies, market forces, and external factors. Countries strive to strike a balance that promotes growth, trade, and financial stability while avoiding excessive exchange rate volatility. Here are some key points related to foreign exchange stability and its related issues:

- a) **Exchange Rate Stability:** Exchange rate stability implies that the value of a country's currency remains steady or experiences only gradual fluctuations about other currencies. This stability can make it easier for businesses, investors, and governments to plan for the future.
- b) **Economic Factors:** Several economic factors can influence exchange rate stability, including inflation, interest rates, trade balances, and fiscal policies. A country with low inflation, stable interest rates, a trade surplus, and responsible fiscal policies is more likely to have a stable exchange rate.
- c) **Central Bank Interventions:** Central banks often play a crucial role in maintaining exchange rate stability. They can buy or sell their currency in the foreign exchange market to influence its value. For example, if a country's currency is appreciating too rapidly, the central bank may intervene by selling its currency to increase its supply and decrease its value.
- d) **Fixed vs. Floating Exchange Rates:** Countries can adopt different exchange rate systems. Some use fixed or pegged exchange rates, where the value of their currency is tied to a specific foreign currency (e.g., the US dollar). Others use floating exchange rates, which are determined by supply and demand in the foreign exchange market. Maintaining stability can be more challenging for countries with fixed exchange rates.
- e) **Speculation and Market Sentiment:** Exchange rates are also influenced by speculation and market sentiment. Traders and investors make decisions based on their expectations of future exchange rate movements, which can sometimes lead to sharp and unpredictable fluctuations.

- f) **External Shocks:** Events such as financial crises, geopolitical tensions including the war, or natural disasters can disrupt exchange rate stability. These external shocks may lead to sudden and significant currency depreciation or appreciation.
- g) **Currency Reserves:** Accumulating foreign exchange reserves can help a country maintain exchange rate stability. These reserves can be used to intervene in the foreign exchange market when necessary. However, building and maintaining sufficient reserves can be expensive.
- h) **International Cooperation:** Exchange rate stability is often a subject of international cooperation, especially within organizations like the International Monetary Fund (IMF). Countries may seek financial assistance or policy coordination to address exchange rate-related challenges.
- i) **Challenges of Overvaluation and Undervaluation:** An overvalued currency (one that is too strong) can hurt a country's exports, while an undervalued currency (one that is too weak) can lead to inflation and discourage foreign investment. Striking the right balance is essential for stability.

To maintain the foreign exchange stability, the following measures can be adopted.

- a) **Central Bank Intervention:** Central banks buy/sell currency to control value.
- b) **Fixed Exchange Rate:** Currency pegged to a stable currency, needs intervention.
- c) **Managed Float:** Exchange rates allowed to fluctuate within a range, intervened when extreme.
- d) **Forward Contracts/Options:** Used for hedging currency risk, and stabilizing rates.
- e) **Fiscal/Monetary Policies:** Adjusting interest rates influences currency attractiveness.
- f) **Economic Reforms:** Structural changes boost currency confidence.
- g) **Exchange Rate Bands:** Allow fluctuation within bounds, central bank intervenes at limits.
- h) **Transparency/Communication:** Clear policies reduce uncertainty.
- i) **Foreign Exchange Reserves:** Buffer for currency stabilization, costly to maintain.
- j) **International Cooperation:** Collaborative efforts for exchange rate stability.

28.6 KEY WORDS

Foreign exchange stability	: Consistency and predictability of exchange rates between different currencies.
Direct quote	: It measures how many units of domestic currency can be bought by one unit of a foreign currency.
Indirect quote	: It measures how many units of foreign currency can be bought by one unit of domestic currency.
Fixed Exchange Rate	: Under this regime, a country's currency is pegged to another currency or a fixed value like gold.
Floating Exchange Rate	: In this regime, the exchange rate is determined by market forces, such as supply and demand.
Managed float	: is a regime for exchange rates where the exchange rate is neither fully set nor completely free.

28.7 SOME USEFUL BOOKS

Case, K. E. (2012). *Principles of Macroeconomics*. Prentice Hall.

Krugman, P. R. (2012). *International Economics: Theory & Policy* (9th ed.). Pearson Addison-Wesley.

Mankiw, N. G. (2009). *Macroeconomics* (7th ed.). Worth Publishers.

Stutz, F. P., & Warf, B. (2010). *The World Economy: Geography, Business, Development*. Pearson Education, Limited.

28.8 ANSWER / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) It affects the cost of imports and exports, the competitiveness of industries, and the overall economic health of a country.
- 2) Receipts of exports, repayment of international loans, arbitrage purpose, speculation, gifts and grants from foreign countries, investments from foreign countries, and foreign remittances.
- 3) Refer to Sub-section 28.2.2

Check Your Progress 2

- 1) $\bar{S} - I(r^*) = NX$. Refer to section 28.3 for details.

- 2) Net export is a negative function of the real exchange rate. Symbolically, $NX = f(\theta)$
- 3) Managed or Dirty Floating, Exchange Rate Policy, Parallel Foreign Exchange Systems, Political Instability, Trade Balances, Inflation and Interest Rates, Global Events.
- 4) Central Bank Intervention, fixed Exchange Rate Systems, managed float, fiscal and monetary policies, economic reforms, exchange rate bands, transparency and communication, foreign exchange reserves, and international cooperation.



UNIT 29 BEHAVIOURAL FINANCE

Structure

- 29.0 Objectives
- 29.1 Introduction
- 29.2 Concept of Behavioural Finance
- 29.3 Difference between Traditional Finance and Behavioural Finance
- 29.4 Growth and Origin of Behavioural Finance
- 29.5 Efficient Markets Hypothesis and Anomalies
 - 29.5.1 Efficient Market Hypothesis and Its Critique
 - 29.5.2 Market Anomalies
 - 29.5.3 Market Puzzles and Behavioural Explanation
- 29.6 Irrational Investor: Cognitive, Social and Emotional Influences on the Investor
 - 29.6.1 Heuristics
 - 29.6.2 Sources of Irrational Investor Behaviour
 - 29.6.2.1 Cognitive Influences on Investor Behaviour
 - 29.6.2.2 Social Influences on Investor Behaviour
 - 29.6.2.3 Emotional Influences on Investor Behaviour
- 29.7 Let Us Sum Up
- 29.8 Key Words
- 29.9 Some Useful Books
- 29.10 Answers / Hints to Check Your Progress Exercises
- 29.11 Terminal Questions

29.0 OBJECTIVES

After studying this unit, you will be able to:

- explain the concept of behavioural finance;
- describe the departure from traditional finance concepts to behavioural finance;
- differentiate between traditional finance and behavioural finance;
- explain the Efficient Market Hypothesis;
- give a behavioural explanation to important market puzzles;
- explain the causes of irrational investor behaviour; and
- describe the heuristics and biases that explain it.

29.1 INTRODUCTION

Traditionally, finance has been based around the concept of “efficient markets” which states that the price of an asset or security is ‘right’, i.e., the price reflects its fundamental value and all the available information. For a long time, financial theory was governed by the idea that investors act rationally as they have all the available information and they aim to maximize profits. The theory believes that even if there is any ‘noise’ by irrational investors in the market, it is balanced by rational investors. Traditional finance has been surrounded by three unbounded traits (rationality, self-control and self-interest) which form a base for ‘homo economicus’.

However, over time, it was felt that the traditional theory was not able to explain various market anomalies like market volatility and unpredictability in the market, market crashes, irrational investor behaviour, stock market bubbles etc. Traditional finance also was not able to address questions like ‘why does an investor trade, how does an investor trade, what are the factors that help an investor decide his portfolio etc.

These market anomalies and investor behaviour were explained by an alternative approach termed ‘Behavioural Finance’ which focuses on cognitive, sociological and psychological influences on the decision-making of investors and financial markets. It focuses on other important aspects of the decision-making process, i.e., how people invest, what they value and how they adjust for risk by embracing findings from psychology and sociology. Behavioural finance highlights that the investors are not always rational and the prices are not always efficient.

Over the last few decades, behavioural finance has become an extremely popular field of study as an application in business and finance. It centres around the fact that investors’ decisions are influenced by their own biases. Thus, investors behave quite differently from ‘homo economicus’ and their financial choices are guided by their social and psychological needs. It also highlights that investors as humans have limits to self-control and often apply heuristics to make complex decisions.

We have discussed traditional finance and the ‘Efficient Market Hypothesis’ (EMH) in previous units. In the present unit, we will discuss what is beyond traditional finance and EMH. In this unit, we will discuss behavioural finance, the factors that led to the growth of this parallel field and how it attempts to explain investors’ behaviour and financial markets.

29.2 CONCEPT OF BEHAVIOURAL FINANCE

As discussed in the previous section, behavioural finance focuses on understanding the thought process of investors and how it impacts their decisions. It attempts to explain the degree of various influences on a

person's decision-making. It further explains various market anomalies and heuristics and biases that cause it.

Behavioural finance became popular after 2002 when Daniel Kahneman was awarded the Nobel Prize for his work on psychological research into economic science, especially concerning human judgment and decision-making under uncertainty. Behavioural finance is studied in two aspects: individual investors, and the whole market. Accordingly, behavioural finance is classified into two aspects: (i) microbehavioural finance, and (ii) macrobehavioural finance.

Microbehavioural finance deals with the cognitive, psychological and sociological influences on the individual investors. It delves into behavioural biases that distinguish them from rational economic beings. It also studies heuristics as rules of thumb that lead to suboptimal decision-making.

Macro-behavioural finance, on the other hand, examines market anomalies that make the market inefficient. It focuses on the reasons behind stock bubbles, market volatility and limited arbitrage. This unit deals with both aspects of behavioural finance.

Check Your Progress 1

1) What is meant by behavioural finance?

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2) What are the assumptions of traditional finance theory?

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3) Which one of the following is not a trait of 'homo-economicus'?

- a) Rationality
- b) Self-control
- c) Cognitive bias
- d) Self-interest

4) Microbehavioural finance deals with:

- a) Irrational investor Behaviour
- b) Stock market volatility
- c) Bubbles
- d) Limited arbitrage

29.3 DIFFERENCE BETWEEN TRADITIONAL FINANCE AND BEHAVIOURAL FINANCE

Traditional finance centres around the concept of investor and market rationality. It suggests that given the available information, investors make financial decisions and in the course of decision-making, are not affected by emotions. Traditional finance does not consider the role of heuristics and accordingly does not contemplate investors being affected by any bias. On the other hand, behavioural finance examines the role of psychology and sociology in financial decision-making. It considers the influence that mood, fear, overconfidence, past experiences etc. have on investors when they decide to devote funds to any asset.

More than contrasting, behavioural finance provides an alternative approach to traditional finance to have a better understanding of the market and investors. Behavioural finance draws from real life that investors are sometimes irrational, and biased and are influenced by various social and psychological factors. For example, it is often observed that while buying stocks, many times investors follow other investors because they think that everyone else knows more and makes better decisions. This reflects a herding mentality and can lead to sub-optimal decisions by an investor.

The main points of contrast between traditional and behavioural finance are:

- Traditional finance centres around the idea of investor rationality and an unbiased approach to processing information for decision-making.

While behavioural finance considers the influence of various biases in investment decision-making, it recognizes that human emotions and other psychological influences may make an investor irrational. Overconfidence, familiarity bias, herding mentality, etc. deter the decision-making capacity of an investor.

- Traditional finance assumes the availability of complete information to everyone in the market. It further assumes that investors examine all the information rationally before making a decision. Thus, investors always make the right decision. Behavioural finance, on the other hand, centres around the idea of bounded rationality, i.e., investors do not process all the information and they try to satisfice instead of maximise. Even if the information available is accurate, investors will still make errors in judgment.
- Traditional finance centres around the 'efficient markets hypothesis', i.e., markets are efficient and the price of a stock always reflects its fundamental value. It is based on the availability of complete knowledge and investor rationality. However, behavioural finance disagrees with the fact that markets are efficient and discusses various anomalies that lead to market volatility and cannot be explained through EMH.

- Investors, according to traditional finance have perfect self-control and they do not let any other aspect of their past or personality influence their decision-making. Behavioural finance considers self-control problems in investors and examines how it affects decision making in turn affecting the whole market.
- Classical finance theory asserts that any noise is balanced out by rational traders but according to behavioural finance, there is a limit to arbitrage and prices are not always right and may stay in a non-equilibrium state for certain periods.

29.4 GROWTH AND ORIGIN OF BEHAVIOURAL FINANCE

The cognitive and psychological underpinnings to explain stock market Behaviour can be traced back to the 60s and 70s also but it was mainly in the 1980s when Behavioural finance evolved with an attempt to explain stock market bubbles and other market anomalies. Since then, foundational ideas of Behavioural Economics have been used to explain the reason behind these anomalies. The ideas of Daniel Kahneman and Amos Tversky have played a major role in explaining the contradictions in the traditional theory of finance.

Over time, behavioural finance has developed through two phases: (i) through examples of market anomalies and deviation from standard finance assumptions; and (ii) through explanation of these anomalies. The decade of the 1980s saw various important concepts like ‘disposition effect’ and ‘equity premium puzzle’ being developed by various economists. In 1981, Robert James Shiller discussed market anomalies and pointed out how rational factors alone cannot explain stock market volatility. De Bondt and Richard Thaler’s paper on stock market overreaction in 1985 elaborated on value anomaly. We will discuss all these concepts later in the unit.

The late 1990s and the early 2000s saw behavioural finance becoming part of mainstream finance. It was able to explain the reasons behind many market anomalies. Many concepts were developed during this time such as (i) limit to arbitrage, (ii) myopic loss aversion, (iii) over-reactions, and under-reactions of the market. This decade also saw research in the area of irrational investors and the behavioural mistakes they make. Robert Shiller's famous book ‘Irrational Exuberance’ in 2000 rightly predicted the crash of the dot-com stock market bubble and spread the ideas of behavioural finance as a parallel field of traditional finance, an understanding of which could increase the predictability of the stock market. Around the same time, Andrei Shleifer also worked to explain how noise traders in the financial market lead to limited arbitrage and cause market anomalies. Further, the Nobel Prize for two consecutive years (2001 and 2002) increased the popularity and acceptance of behavioural finance. You may be aware that the Nobel Prize was awarded in 2001 to George Akerlof, Michael Spence, and Joseph Stiglitz

“for their analyses of markets with asymmetric information” and to Daniel Kahneman in 2002 for “having integrated insights from psychological research into economic science, especially concerning human judgment and decision-making under uncertainty”.

As of now, the field of behavioural finance is still evolving and is a popular research area. The psychological, cognitive and sociological foundations of financial decision-making are still being explored to have a better understanding of the markets. However, behavioural finance is criticized for being “Micro-efficient” and not “Macro-inefficient”, i.e., it can explain individual behaviour of investors effectively but is not very useful in predicting the overall market. We can explain market anomalies through behavioural finance in the long run, but if we try to predict variations in the stock market in the short run we will not be very successful. The field is still growing, with more and more research to understand sources of irrational behaviour and market anomalies.

Check Your Progress 2

- 1) Compare and contrast traditional finance with behavioural finance.

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- 2) Discuss the growth and evolution of behavioural finance.

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- 3) Traditional theory assumes that investors are _____
whereas behavioural finance considers investors to be _____

- a) Rational, rational
- b) rational, irrational
- c) irrational, irrational
- d) irrational, rational

- 4) Which one of the following is not true about behavioural finance?
- a) It assumes the availability of complete information to everyone in the market.
 - b) It considers issues such as self-control in investors and examines how it affects decision making which in turn affects the whole market.
 - c) it centres around the idea of bounded rationality.
 - d) There is a limit to arbitrage, and prices are not always right and may stay in a non-equilibrium state for certain periods.

29.5 EFFICIENT MARKETS HYPOTHESIS AND ANOMALIES

29.5.1 Efficient Market Hypothesis and its Critique

We have already discussed the concept of ‘efficient markets hypotheses’ (EMH) in Unit 25. According to EMH, market prices fully reflect all available information. The EMH is one of the important cornerstones of traditional finance theories. It was developed independently by Paul A. Samuelson and Eugene F. Fama in the 1960s and has been applied extensively in traditional finance. In an efficient market, all relevant information is freely available to profit-maximizing rational investors who are trying to predict the value of various securities. According to EMH, markets are rational and prices of stocks fully reflect all available information. Since such information is readily available, the price of securities quickly adjusts to the change. The idea behind EMH is that when some new information arises or any event occurs, the news spreads very quickly and is incorporated into stock prices. For example, if a currency note is lying on the road, it will not be there for long as it will be surely picked up by someone. Similarly, all information is picked up by market agents and used in decision-making. Current prices are considered to be the best approximation of a company’s intrinsic value.

The EMH is associated with the idea of a “random walk”. According to the random walk theory, past movements or trends of a stock price (or financial market) cannot be used to predict its future movement and it is not possible to outperform the market without assuming additional risk. The logic behind the random walk theory is that stock prices reflect information or news about the stocks. Since the news is unpredictable, today’s situation or price cannot be used to predict future values. As such, experienced investors do not have any added advantage over uninformed investors.

Behavioural finance, on the other hand, believed in ‘momentum’ in short-run stock prices. According to behaviour finance, investors follow market trends, i.e., if they notice a rise in the price of a stock, they invest in the market in a

kind of “bandwagon effect.” leading to sometimes what is called ‘irrational exuberance’, i.e., investor enthusiasm that leads to rising in a stock price higher than its fundamental value. Another rationale for ‘momentum’ as given by behavioural finance is the investor’s under-reaction to the new information. Investors take time to change their existing perception or understanding of a particular stock. Thus, the full impact of any news or announcement is realized over a period leading to a positive correlation in prices (thus, changes in prices are not random)

Another significant critique of behavioural finance was regarding assumptions about human Behaviour. They critiqued the rationality assumption by classical finance and focused on sociological, psychological and cognitive influences leading to behavioural anomalies. Behavioural finance documented various departures from a classical paradigm in the form of behavioural biases that guide investor behaviour especially while deciding under uncertainty. Many times, these biases lead to irrational behaviour and ultimately to sub-optimal outcomes in the market. Some of the biases that we will discuss in this unit are herding, overconfidence, overreaction, loss aversion, and mental accounting which dominate the thinking process of an investor. These biases in turn lead to market anomalies such as the ‘equity premium puzzle’ (Stocks have earned disproportionately higher returns over the years yet investors are relatively unwilling to hold them), the ‘volatility puzzle’ (Stock returns are highly variable with large dispersions) and the ‘predictability puzzle’ (Stock returns are predictable suggesting that persistent profits are not being eroded by arbitrage).

29.5.2 Market Anomalies

Anomalies are occurrences that deviate from the predictions of the efficient markets hypothesis and undermine its core assumptions. (Tversky & Kahneman 1986) defined market anomalies as “an anomaly is a deviation from the presently accepted paradigms that are too widespread to be ignored, too systematic to be dismissed as random error and too fundamental to be accommodated by relaxing the normative system”.

Anomalies can be of three types: (i) Fundamental Anomalies, (ii) Technical Anomalies, and (iii) Calendar or Seasonal Anomalies

i) Fundamental Anomalies

- Value Anomaly: It refers to the over-estimation of future earnings and returns of growth companies.
- Low Price to Book: It is often noticed in the stock market that the stocks having a high book-to-market ratio generate less return than the stocks with a low price-to-book ratio.
- High Dividend Yield: There is a positive correlation between yield and stock return. Stocks with high dividend yields outperform the market and generate more returns.

- **Neglected Stocks:** The prior neglected stocks generate more return subsequently over some time. Accordingly, the prior best performers underperform the index.
- **Low Price to Earnings (P/E):** The stocks with low price-to-earnings ratios are likely to generate more returns and outperform the market, while the stocks with high price-to-earnings ratios tend to underperform in the market.

ii) Calendar Anomalies

Calendar anomalies reflect an irregular pattern of stock returns in various periods, i.e., movement in stock prices from day to day, month to month, year to year etc. Some of the Calendar anomalies are as follows:

- **Weekend Effect:** The stock prices are likely to fall on Monday, i.e., the closing price of Monday is less than the closing price of the previous Friday.
- **Turn-of-the-Month Effect:** It reflects the tendency of stock prices to increase in the last trading day of the following month, and the first three days of next month.
- **Turn-of-the-Year Effect:** This anomaly describes the increase in the prices of stocks and trading volume of the stock exchange in the last week of December and the first half month of January.
- **January Effect:** The tendency of small-company stocks to generate more return than another asset in the first two to three weeks of January is called the January effect.

iii) Technical Anomalies

Various techniques can be used to forecast future stock prices based on past prices and relevant information in the market. Traditionally, researchers asserted that the market holds weak form efficiency, prices already reflect all the information and everyone has all the information. Thus, an investor cannot beat the market by earning abnormal returns. But various market anomalies deviate from the traditional viewpoint:

- **Moving Averages:** Buying and selling signals of stocks can be generated by long-period averages and short-period averages. A purchase signal is created when short-period averages rise over long-period averages. The best time to sell is when short-period averages fall below the long-period averages.
- **Trading Range Break:** A breakout is a potential trading opportunity that investors apply to make huge profits. This technique involves identifying support and resistance levels along with current price trends to identify possible entry and exit points. By buying or selling at the right time, investors can make huge profits.

29.5.3 Market Puzzles and Behavioural Explanation

Over time, Behavioural Finance has attempted to explain various market anomalies and puzzles. This section discusses about few popular puzzles in the stock market and their Behavioural explanation.

i) Equity premium puzzle

The equity premium puzzle, first documented by Mehra and Prescott, refers to the empirical fact that stocks have greatly outperformed bonds over the last century. Stocks have earned disproportionately higher returns than bonds, however, people still buy bonds. The level of risk aversion required for such a choice by investors has to be very high. Thus, the equity premium puzzle cannot be explained in terms of risk aversion. Behavioural finance explains this puzzle in terms of 'Prospect Theory'. The first explanation is provided in terms of 'loss aversion', i.e., investors are more sensitive to losses than to gains. Secondly, investors are myopic, i.e., investors are assumed to evaluate their portfolio frequently even if they are saving for their retirement. Thus, myopic loss aversion explains the equity premium puzzle. Another explanation for the puzzle is provided through 'ambiguity aversion'. Ambiguity aversion is a preference for known risks over unknown risks. Since the probability distribution of equity returns is unknown, people overweigh the probability of returns from bonds.

ii) Volatility Puzzle

The volatility puzzle reflects the high variability of stock prices having large dispersions. It is observed in the stock market that changes in stock prices are disproportionately higher than the changes in fundamentals. This observation is against the theory of EMH according to which, prices always represent the fundamentals of the assets.

Behavioural finance attempts to solve this puzzle through various biases that investors have. Various cognitive, social and emotional biases lead to sub-optimal decision-making by the investors which in turn affect the stock market and stock prices. It is noticed that while making a decision, investors overweigh more recent information, and consensus beliefs and seek confirming evidence which makes them under-react or overreact in the market.

Investors also tend to focus more on similar pieces of information called focal points which also affect their decision making. The reason behind the steep fall in prices rather than the upward correction is that people are loss-averse and tend to overweigh negative information and feedback.

iii) Predictability Puzzle

This puzzle states that stock returns are predictable from price-dividend ratios. Behavioural finance is explained in terms of behavioural biases like the disposition effect (the tendency to prematurely sell assets that have made

financial gains, while holding on to assets that are losing money), momentum effect (the tendency of stocks that performed well in the past months to continue to do well in the following period and vice versa for stocks with a poor performance).

Check Your Progress 3

- 1) List the various market anomalies observed in the market.

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- 2) What is the behavioural explanation of the equity premium puzzle?

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- 3) Explain the volatility puzzle from a behavioural perspective.

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- 4) Which of the following explains the Equity premium puzzle?

- a) Under-reaction
- b) Disposition effect
- c) Momentum effect
- d) Myopic loss aversion

29.6 IRRATIONAL INVESTOR: COGNITIVE, SOCIAL AND EMOTIONAL INFLUENCES ON AN INVESTOR

Behavioural finance replaces the idea of rational decision-makers called 'econs' with 'humans' who are influenced by cognitive, social and emotional factors. Humans often are more biased than logical while making financial decisions. Before we discuss the biases that affect human decision-making, let us discuss how humans make decisions. You must have noticed that some decisions like morning routine, going to the office etc. are automatic but some decisions like financial investments etc. require attention, focus and analysis. In real life, decision-making is a complex process and is influenced by context, cues, social norms and past experiences.

When faced with difficult situations, people use heuristics or ‘rules of thumb’ to make decisions. These rules are imperfect but functional shortcuts that help investors simplify complex situations and make quick decisions.

The concept of heuristics was first introduced by the economist and cognitive psychologist Herbert A. Simon in the 1950s. He used this approach to answer the question of how humans make decisions when the conditions for rational choice theory are not met, that is how people decide under uncertainty. He suggested that there were limitations to rational decision-making.

Later, in the 1970s, psychologists Amos Tversky and Daniel Kahneman worked on ‘Heuristics’ to understand cognitive biases. Heuristics have since been used to explain the causes of irrational investors and why people’s intuitive judgement deviates from the rule. Kahneman has discussed how heuristics can be understood in terms of substitution, i.e., when faced with a difficult problem, the one that individuals are not able to address directly, they replace it with an easier question and answer that instead. For example: Instead of addressing the question: “How much profit this stock is going to make”, the investor will substitute the question: “What do I feel about this stock?” Substitution helps to provide quick answers but may lead to systematic and predictable errors called biases. In the above example, the investor has used the affect heuristic.

29.6.1 Heuristics

Four prominent heuristics are as follows:

i) The anchoring and adjustment heuristic

According to the anchoring and adjustment heuristic, people employ a certain starting point (“the anchor”) and make adjustments until they reach an acceptable value over time. Anchoring is a cognitive bias where investors rely on facts provided before a decision or an estimation is made. The facts may be completely unrelated, but research shows that they have a significant impact on the decision.

Anchoring heuristic is used by people when they have to estimate a value with an unknown attribute. In such cases, people use some initial “anchor” or default number which is then adjusted up or down based on the information available. Approximations or final values derived by a person using heuristics are often biased because anchors can never be adjusted sufficiently. Anchoring and adjustment bias occurs when an investor gives undue prominence to anchors which are most of the times statistically irrelevant. Anchoring occurs to reduce the amount of cognitive load placed on our brains.

For example: If you are asked to estimate the value of Reliance’s stock after 4 months. Since the problem is difficult and there is uncertainty, most people will apply anchoring and adjustment heuristics to find a quick response. They would use the present value of the stock as an

anchor and will adjust up or down based on any other information that they have. This may lead to a biased answer as an investor is looking at information from a twisted lens, i.e., with the present value of the stock in mind.

Property dealers often use ‘anchors’ to start negotiations. They quote higher prices for the property to set an anchor and bargain the deal to give satisfaction to the customer who is influenced by the initial anchor.

ii) The Representativeness Heuristic

According to Kahneman and Tversky (1972), representativeness means, that in situations of uncertainty, people “evaluate the probability of an uncertain event, or sample, by the degree to which it is: (i) similar in essential properties to its parent population; and (ii) reflects the salient features of the process by which it is generated”. Often people choose options or make decisions based on representative information rather than logical and probabilistic reasoning. This heuristic causes judgment errors such as neglecting base rate (people tend to ignore general information or base rate when specific information is also provided), conjunction fallacy (people assume that certain specific conditions are more likely than general conditions) and applying the law of small numbers (exaggeration of the degree to which a small sample will resemble the population from which they are drawn).

In the stock markets, investors often see the past performance of firms as being representative of the firm’s future performance as well. Investors are also seen to overweight the recent information to the detriment of the past information. Thus, if a firm has been making huge profits for a few years, investors might assume that the company will continue to perform great in future also. Representative heuristic sometimes leads to stock market overreactions and underreactions challenging the efficient market hypothesis.

iii) The Availability Heuristics

In complex and uncertain situations, the brain tries to take a mental shortcut to ease the decision-making process. These predictable shortcuts are based on our past experiences and recent memories. The availability heuristic is one such rule of thumb or shortcut. The Availability Heuristic assess the probability of occurrence of the event based on the ease with which it comes to the mind. It is governed by the principle that “if you can think of it, it must be important.” Individuals believe that the things they can recall more easily are more common and more accurate representations of the real world. For instance, the ease with which you can recall theft in your locality will determine how will you rate the law and order situation in your locality. Though a recent crime may not be a true representation of a law-and-order situation in your locality you will overestimate the occurrence of such events because of the availability

heuristic. This may lead to a cognitive bias as a recent event may not be the best representation of reality.

Availability heuristic is widely seen in financial markets also. Recent market news or events affect the investment decision more than the fundamentals of the company. Many times, this leads to an overreaction in the market which in turn might cause crashes or bubbles. In the case of a significant event, investors overestimate the probability of the occurrence of a similar event.

One example of an Availability heuristic leading to bias is the case of 'hot hand' where it is assumed that a person having a string of successes will continue with the string and will be successful. This bias was first noticed in the game of basketball (hence, the name 'hot hand') whereby the players who score maximum baskets keep on getting maximum passes even though they may be just average players. This bias is seen in the case of unrelated events like the roll of a dice or the flipping of a coin.

In financial markets, investors also experience the bias of 'hot hand' when they are more likely to deal with traders who have recently performed exceptionally well in the market. In reality, there is no relationship between past performance and future performance.

iv) The Affect Heuristic

Affect heuristic helps individuals to assign the probabilities of occurrence of an event based on how a person feels about it. If a person feels good about it, a higher probability is assigned to good consequences and vice versa.

Affect heuristic lets the decisions affected by emotions and mood while making a decision. People tend to make more risky decisions when they are happy or in a positive state of mind.

The affect heuristic is most prominent when people do not have the resources or time to reflect.

It is a bias in which emotions — fear, anxiety, surprise, pleasure — influence and quicken a person's decision-making process.

The affect heuristic may lead to overconfidence bias in an investor. Investors may make sub-optimal decisions when emotions override logic or facts.

For example: sometimes investors invest in a company's stock because "they have a good feeling about it". Good feelings may not have anything to do with the fundamentals of the company and thus, may lead to irrational decisions.

As discussed in the previous sections, Behavioural finance postulates that investors are not completely rational, preference-consistent or unaffected by emotions. Investors are boundedly rational and influenced by cognitive biases, social factors and emotions. In this section, we will discuss cognitive, social and emotional influences on an investor.

29.6.2.1 Cognitive Influences on Investor Behaviour

i) Over-Confidence

Often it is seen in financial markets that people tend to attribute success to their ability and failures to external factors. This is called self-attribution bias. This makes them overconfident about their capability which in turn leads to poor decisions regarding stocks. Overconfidence also leads to excessive trading as people having too much confidence in themselves buy too often for which they also have to sell too often (to free up capital).

Another source of overconfidence in investors is hindsight bias. Hindsight bias makes people believe after an event that they knew the outcome of the event before it happened. For example: after a stock market crash, many investors claim that they knew it was coming. When an investor believes that he has exceptional foresight or intuition, it makes him overconfident and more likely to take uncalculated risks.

ii) Cognitive Dissonance

Festinger's theory of cognitive dissonance states that people feel internal tension and anxiety when they are subjected to conflicting beliefs. Every individual tries to reduce cognitive dissonance either by changing past values or feelings or by trying to justify the decision. This is also applicable to financial market investors who attempt to justify wrong investment or decision. Cognitive dissonance leads to irrational Behaviour of selling losers too late as they are not able to accept the fact that they made a wrong decision.

This theory may apply to investors or traders in the stock market who attempt to rationalize contradictory behaviour. Investors sell losers too late as they are not willing to accept that they made a wrong decision. They also try to avoid feelings of regret or embarrassment of reporting a loss as a result of poor investment decisions.

iii) Under-reaction and Overreaction in the Market

Investors are often biased in perceiving any new information relative to their prior beliefs. Thus, if any new information or news is against their existing beliefs, they underweigh it and initially underreact to it. In the weeks following the information, they start to adjust their investments.

Similarly, when any new information, news or announcements are similar to their existing beliefs, investors overreact to it. Both underreaction and overreaction are irrational investor Behaviour and lead to sub-optimal decision-making. This leads to stock market volatility.

iv) Disposition Effects

The disposition effect refers to the tendency to prematurely sell assets that have made financial gains while holding on to assets that are losing money. Investors sell profitable investments to make quick profits while holding on to losing investments in the hope of converting them into gains. The disposition effect reflects irrational behaviour as investors lose out on possible gains due to momentum when they sell winning assets too quickly.

Daniel Kahneman has explained the disposition effect through Prospect theory. According to the theory, investors become risk-averse after gains and choose to realise the gain rather than risk losing it. On the other hand, investors become risk-seeking after a loss and hold on to the risky asset.

v) Diversification Bias

People seek more variety when they choose multiple items for future consumption simultaneously than when they make choices sequentially, i.e., on an 'in the moment' basis. Diversification is non-optimal when people overestimate their need for diversity. When they have many options, they diversify even if it is in their best interest to invest in a single asset.

On the other hand, sometimes investors diversify insufficiently. Availability heuristics and familiarity biases lead to heavy investment towards companies from investors' home countries. There is no economic justification for the home bias but familiarity reduces the perception of risks.

vi) Mental Accounting

Mental accounting refers to a set of cognitive operations that investors use to keep track of their financial investments. Investors classify their money differently based on various subjective criteria. Investors invest in both safe and speculative assets and keep these investments in different mental accounts so that negative returns from speculative assets cannot affect the positive returns from safe assets. Thus, investors keep money that they can afford to lose for speculative purposes. Rationally, there should be no division between safety capital and money that an individual can afford to lose. This mental bias often leads people to make irrational decisions and financially counterproductive investment decisions.

i) Herding

Herding in financial markets generates speculative bubbles when traders are tracking the decisions of others rather than the fundamental value of assets.

Investors with herding instincts follow others and make similar investments rather than trusting their analysis. Herd instinct can lead to Asset bubbles, panic buying or selling and can be very detrimental to the market.

Herding is the outcome of social learning when people believe that others are better informed and thus follow the crowd. During financial speculation, it is seen that investors buy at exorbitantly high prices not because they think that the asset is worth it but because they believe that others think it is.

ii) Reputation

Herding often emerges as a reputation concern as well. Keynes postulated that it is rational and follow the crowd and herd as it helps to maintain a good reputation. People believe that it is better to fail conventionally than to succeed unconventionally

iii) Beauty contests

Financial beauty contests were first used by Keynes to describe the second-guessing that characterizes financial speculation. The beauty contest is a newspaper competition where competitors are asked to select from a series of photos which according to them will be liked by the majority of people. It is similar to iterated reasoning used in financial speculation. Speculators are interested in identifying short-term investment plans of others and not what they think will be the best investment.

Financial speculations subject asset prices to financial loops which in turn might lead to financial instability. Instability is magnified if borrowings are the source of asset funding.

iv) Success Stories and Advice from Family

Investors are also affected by success stories and advice from the family. It was found in various research studies that people tend to invest more when their neighbours and family members have made profits from investing in the stock market. Interestingly, it was also found that investors are not negatively influenced by the failures of people around them mainly because people share success stories more than failures.

v) Expert Influence

Investors are also influenced by stock market experts. Analysts many times have incentives to tout particular shares or are influenced by other analysts or previously released information. This leads to herding and inferior returns in the market.

29.6.2.3 Emotional Influences on Investors

i) Feelings of Regret

Fear of regret deals with the emotional reaction that people experience after realizing that they have made a mistake or error in making a decision. When an investment goes bad, people act irrationally by trying to avoid selling it to avoid feelings of regret. Sometimes investors avoid the possibility of regret by following what other investors are doing. They feel less embarrassed about losing money in popular investments than in unpopular investments. The theory of Regret can be explained in terms of cognitive dissonance which we have already discussed under 'Cognitive influences on investor Behaviour'.

ii) Loss Aversion

Investors find losses to be disproportionately painful as compared to the pleasure from the gains. Loss aversion makes people prefer things as they are. People will overvalue things that they already own leading to divergences between willingness to pay and willingness to accept. People are willing to pay less for an object that they do not yet own than they will accept when selling the same object that they already own. For example: If you own an antique art piece worth a market price of Rs. 50,000, you will not be willing to pay Rs. 50,000 for it at this point but would expect a much higher price for it if you sell it.

iii) Sunk Cost Fallacy

The sunk cost fallacy describes an emotional tendency to invest more money, time, and effort into a project where we have already invested even if a person knows that the investment is going to fail.

The sunk cost fallacy is closely linked to 'loss aversion'. Investors become emotionally involved in a stock or project that they are not willing to accept the failure. Thus, they invest more money to make that investment work. People have difficulty in letting go of the sunk cost and they make additional mistakes like:

- **Aggressive Investing:** People who were risk averse initially become more aggressive in investing after losing a certain amount of money. They start taking undue risks which might lead to huge losses.
- **Averaging:** People invest more in the stocks where they have lost money as they try to average out returns in those stocks. This may lead to even more losses.

1) Explain various cognitive influences on an investor.

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2) What is the sunk cost fallacy?

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3) Consider the following statements. Name the bias/effect that an investor suffers from.

- a) An investor believes that others are better informed and thus follows them _____
- b) An investor attributes success to his ability and failures to external factors. _____
- c) An investor sells assets that have made financial gains while holding on to assets that are losing money. _____
- d) An investor finds losses to be disproportionately painful as compared to the pleasure from the gains. _____

29.7 LET US SUM UP

Traditional finance has been based on the concept of “efficient markets” which states that the price of an asset or security is ‘right’ as it reflects its fundamental value. Financial theory was governed by the idea that investors are rational.

However, over time, it was felt that the traditional theory was not able to explain market volatility and unpredictability in the market, market crashes, irrational investor Behaviour, stock market bubbles etc. These were explained by an alternative approach termed ‘Behavioural Finance’ which focuses on cognitive, sociological and psychological influences on the decision-making of investors and financial markets. More than contrasting, behavioural finance provides an alternative approach to traditional finance to have a better understanding of the market and investors. Behavioural finance draws from real life that investors are sometimes irrational, and biased and are influenced

by various social and psychological factors. It evolved mainly in the 1980s when Behavioural finance evolved with an attempt to explain stock market bubbles and other market anomalies.

In this unit, we also discussed about efficient markets hypothesis and its anomalies. According to EMH, markets are rational and prices of stocks fully reflect all available information and since the information is readily available, the price of securities quickly adjusts to the change. There are fundamental, technical and Calendar or season and all anomalies. These anomalies reflect occurrences that deviate from the predictions of the efficient markets hypothesis and undermine its core assumptions. Behavioural Finance also explains various market puzzles. The equity premium puzzle refers to the empirical fact that stocks have greatly outperformed bonds over the last century. Stocks have earned disproportionately higher returns than bonds, however, people still buy bonds. Behavioural finance explains this puzzle through the concepts of myopic loss aversion and Ambiguity aversion. The volatility puzzle reflects the high variability of stock prices having large dispersions. Behavioural finance attempts to solve this puzzle through various cognitive, social and emotional biases that investors have. This puzzle states that stock returns are predictable from price-dividend ratios. Behavioural finance is explained in terms of behavioural biases like the disposition effect, momentum effect and post-earning announcement drift.

This unit also discussed about four prominent heuristics that people follow while making decisions. Heuristics or ‘rules of thumb’ are imperfect but functional shortcuts that help investors simplify complex situations and make quick decisions. These heuristics are The anchoring and adjustment heuristic (people employ a certain starting point (“the anchor”) and make adjustments until they reach an acceptable value over time), the Representativeness heuristic (In situations of uncertainty, people evaluate the probability of an uncertain event by comparing it to an existing prototype that already exists in our minds), Availability Heuristics (assess the probability of occurrence of the event based on the ease with which it comes to the mind), Affect Heuristic (helps an individual to assign probabilities of occurrence of an event based on how a person feel about it).

We also discussed cognitive influences (Over-Confidence, Cognitive Dissonance, Under-reaction and Over-reaction, Disposition effects, Diversification bias, Mental Accounting), social influences (Herding, Reputation, Beauty contests, Success stories and advice from family, expert influence), and emotional influences (Feelings of Regret, Loss aversion, Sunk cost fallacy) on an investor.

29.8 KEY WORDS

Affect Heuristic : Affect heuristic helps individuals to make decisions based on how a person feels about it.

Ambiguity aversion: Ambiguity aversion is a preference for known over unknown risks by an individual.

Anchoring and adjustment heuristic	: Heuristic whereby a person is heavily dependent on the information received initially (called the “anchor”) and then adjusts it towards the problem while making decisions.
Market Anomaly	: An anomaly is a deviation from the presently accepted paradigms or assumptions about the market which cannot be ignored as a random error.
Availability Heuristics	: It refers to the heuristic that assesses the probability of occurrence of the event based on the ease with which it comes to the mind.
Behavioural Finance	: It focuses on cognitive, sociological and psychological influences on the decision-making of investors and financial markets.
Cognitive Dissonance	: It refers to the internal tension and anxiety that people experience when they are subjected to conflicting beliefs or experiences opposite to their prior beliefs.
Disposition Effects	: It refers to the tendency to prematurely sell assets that have made financial gains while holding on to assets that are losing money.
Efficient Market Hypothesis	: An investment theory argues that markets are efficient, and prices reflect the fundamental value of an asset and thus there is no scope to make excess profits by investing.
Equity premium puzzle	: The equity premium puzzle refers to the inability of traditional finance models to explain the average equity risk premium over bonds.
Herding	: Herding is the outcome of social learning when people believe that others are better informed and thus follow the crowd.
Heuristics	: Heuristics or ‘rules of thumb’ are imperfect but functional shortcuts that help investors simplify complex situations and make quick decisions.
Loss Aversion	: Investors find losses to be more painful as compared to the pleasure from gains. Loss aversion makes people prefer things as they are.
Mental Accounting	: Mental accounting refers to a set of cognitive operations that investors use to classify their money differently in different portfolios based on various subjective criteria.
Predictability Puzzle	: Stock returns are predictable suggesting that persistent profits are not being eroded by arbitrage.

- Random Walk** : It refers to past movements or trends of a stock price or market that cannot be used to predict its future movement and it is not possible to outperform the market without assuming additional risk.
- Representativeness Heuristic** : In situations of uncertainty, people evaluate the probability of an uncertain event by comparing it to an existing prototype that already exists in our minds.
- Sunk Cost Fallacy** : It describes an emotional tendency to invest more money, time, and effort into a project where we have already invested even if a person knows that the investment is going to fail.
- Theory of Regret** : When an investment goes bad, people act irrationally by trying to avoid selling it to avoid feelings of regret.
- Volatility Puzzle** : The volatility puzzle reflects the high variability of stock prices having large dispersions
- Law of One Price** : The law says that identical securities have identical prices because prices are based on fundamental values. The law of one price limits the opportunity to arbitrage.

29.9 SOME USEFUL BOOKS

Baddeley, Michelle (2013). *Behavioural Economics and Finance*. Routledge. Chapter 11: Financial Instability

Ranyard, Rob(2018) *Economic Psychology*. British Psychological Society and Wiley. Chapter 15: Behaviour in Financial Markets.

Tversky, A., & Kahneman, D. (1986). Judgment under uncertainty: Heuristics and biases. In H. R. Arkes & K. R. Hammond (Eds.), *Judgment and decision making: An interdisciplinary reader* (pp. 38–55). Cambridge University Press. (This chapter originally appeared in "Science," 1974, 185, 1124-1131)

29.10 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) It focuses on cognitive, sociological and psychological influences on the decision-making of investors and financial markets.
- 2) In efficient markets, the price reflects its fundamental value and all the available information, that investors act rationally, have self-control and act in self-interest.

- 3) c
- 4) a

Check Your Progress 2

- 1) Traditional finance centres around the idea of investor rationality whereas Behavioural finance considers the influence of various biases in investment decision making.

Traditional finance assumes the availability of complete information, Behavioural finance, on the other hand, centres around the idea of bounded rationality.

Traditional finance centres around the 'Efficient market hypothesis'. Behavioural finance discusses various anomalies that lead to market volatility and cannot be explained through EMH.

Classical finance theory asserts fact that any noise is balanced out by rational traders but behavioural finance asserts that there is a limit to arbitrage and prices are not always right.

- 2) Refer to Section 1.4
- 3) B
- 4) A.

Check Your Progress 3

- 1) These are Fundamental anomalies, Technical anomalies, and Calendar or seasonal anomalies (refer to Sub-section 29.5.2).
- 2) Myopic loss aversion and ambiguity aversion explain the equity premium puzzle.
- 3) Various cognitive, social and emotional biases lead to sub-optimal decision-making by the investors which in turn affect the stock market and stock prices.
- 4) d

Check Your Progress 4

- 1) These are Over-Confidence, Cognitive dissonance, Under-reaction and over-reaction, Disposition effects, Diversification bias, and Mental Accounting (refer to Sub-section 29.6.2 for details).
- 2) The sunk cost fallacy describes an emotional tendency to invest more money, time, and effort into a project where we have already invested even if a person knows that the investment is going to fail.
- 3)
 - a) Herding
 - b) Self-attribution bias
 - c) Disposition effect
 - d) Loss aversion

29.11 TERMINAL QUESTIONS

- 1) How is behavioural finance different from traditional finance?
- 2) Behavioural Finance posits that heuristics help investors to make decisions quickly. Discuss the various heuristics and how are they applied by investors. Elaborate on the biases that these heuristics cause.
- 3) Give behavioural explanations to various market puzzles that are not addressed by traditional finance.
- 4) Describe various anomalies in the Efficient Markets Hypothesis.
- 5) Bring out the cognitive influences on investors that lead to various biases in decision-making.
- 6) Explain social influences on investors' decision-making
- 7) What are the various emotional influences that bias investor decision-making?
- 8) Define:
 - a) Sunk cost fallacy
 - b) Herding
 - c) Loss aversion
 - d) Mental accounting
 - e) Diversification bias
- 9) What is the Theory of Regret? How does it influence decision-making?
- 10) Describe the equity premium puzzle.
- 11) Describe predictability puzzle
- 12) 'Behavioural finance is 'micro efficient' but not macro efficient'. Comment.